var, x, y, z, w
$label$, L, $Next$, $Done$
lit
f
$dsort$, D
$tvar$, a, b
$index$, i, j, k, n

terminals	::=			
	|	$\rightarrow$		
	|	$\rightsquigarrow$		
	|	$\rightarrow$		
	|	$\mid$		
	|	$\vee$		
	|	@		
	|	$\multimap$		
	|	$\wedge$		
	|	$\Rightarrow$		
	|	$\in$		
	|	true		
	|	false		
	|	$\top$		
	|	$\bot$		
	|	$\forall$		
	|			
	|	is		
	|	let		
	|	res		
	|	core		
	|	log		
blit	::=			boolean literals
	|	true		
	|	false		
eff, ϵ	::=			effects
	|	**0**		
	|	**1**		
	|	**S**		
kind	::=			kinds
	|	type		
	|	sort		
τ, A, B, C	::=			sorts
	|	Int		integers
	|	Bool		booleans
	|	Ptr τ'		pointer
	|	$(\tau_1, .., \tau_n)$		tuple
	|	$D(\tau_1, .., \tau_n)$		datatype application
	|	Pred τ'		predicate sort
	|	a		type variable
	|	$[\gamma]\tau$	M	
$\overline{A}$	::=			type lists
	|	$[]$		
	|	$A{::}\overline{A}$		

F	$::=$			function signatures
	$\mid$	$\tau \rightarrow_{\mathit{eff}} \tau'$		
Γ	$::=$			contexts
	$\mid$	$\cdot$		
	$\mid$	$\Gamma, x{:}_{\mathit{eff}}\tau$		
	$\mid$	$\Gamma, a{:}\mathit{kind}$		
	$\mid$	Γ, Γ'		
	$\mid$	$\lvert\Delta\rvert$	M	
γ	$::=$			explicit substitutions
	$\mid$	$\cdot$		
	$\mid$	$\gamma, ce_1/x_1, \ldots, ce_n/x_n$		
	$\mid$	$\gamma, \tau_1/a_1, \ldots, \tau_n/a_n$		
	$\mid$	$\mathsf{id}(\Gamma)$	M	
	$\mid$	$\gamma; \gamma'$	M	
Σ_c	$::=$			core signatures
	$\mid$	$\cdot$		
	$\mid$	$\Sigma_c, f : F$		
	$\mid$	$\Sigma_c, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce$		
	$\mid$	$\Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}$		
	$\mid$	$\Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}$		
Σ_s	$::=$			surface signatures
	$\mid$	$\cdot$		
	$\mid$	$\Sigma_s, f : F$		
	$\mid$	$\Sigma_s, \mathsf{fun}\ f : \tau \rightarrow_{\mathit{eff}} \tau' = \{\overline{\mathbf{pat} \rightarrow \mathbf{se}}\}$		
	$\mid$	$\Sigma_s, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}$		
	$\mid$	$\Sigma_s, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}$		
Σ_r	$::=$			refined signatures
	$\mid$	$\cdot$		
	$\mid$	$\Sigma_r, f : F$		
	$\mid$	$\Sigma_r, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce$		
	$\mid$	$\Sigma_r, f : RF$		
	$\mid$	$\Sigma_r, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}$		
	$\mid$	$\Sigma_r, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}$		
prim	$::=$			primitives
	$\mid$	**Add**		addition
	$\mid$	**Sub**		subtraction
	$\mid$	**Mul**		multiplication
	$\mid$	**Div**		division
	$\mid$	**And**		conjunction
	$\mid$	**Or**		disjunction
	$\mid$	**Not**		negation
	$\mid$	**Lt**		less than
	$\mid$	**Le**		less or equal

	$\mid$	$\mathbf{Gt}$	greater than
	$\mid$	$\mathbf{Ge}$	greater or equal
	$\mid$	$\mathbf{Eq}\,[A]$	equality test
	$\mid$	$\mathbf{New}\,[A]$	allocate
	$\mid$	$\mathbf{Del}\,[A]$	deallocate
	$\mid$	$\mathbf{Get}\,[A]$	dereference
	$\mid$	$\mathbf{Set}\,[A]$	update
	$\mid$	$\mathbf{Own}\,[A]$	ownership
ce	$::=$		core expressions
	$\mid$	x	
	$\mid$	lit	integer literal
	$\mid$	$blit$	boolean literal
	$\mid$	$\mathsf{let}\, x = ce_1; ce_2$	
	$\mid$	$(ce_1, \ldots, ce_n)$	
	$\mid$	$\mathsf{let}\,(x_1, \ldots, x_n) = ce_1; ce_2$	
	$\mid$	$L\, ce$	
	$\mid$	$\mathbf{case}\, ce\, \mathbf{of}\, \{L_1\, x_1 \to ce_1 \mid \ldots \mid L_n\, x_n \to ce_n\}$	
	$\mid$	$\mathbf{iter}\,(x = ce_1)\{ce_2\}$	
	$\mid$	$prim\, ce$	
	$\mid$	$\mathbf{if}\, ce_1\, \mathbf{then}\, ce_2\, \mathbf{else}\, ce_3$	conditional
	$\mid$	$ce : \tau$	
	$\mid$	$f\, ce$	
	$\mid$	$ce_1 \doteq ce_2$	
	$\mid$	$\mathbf{take}\, x = ce_1; ce_2$	
	$\mid$	$\mathbf{return}\, ce$	
	$\mid$	$\mathbf{fail}$	failure
	$\mid$	(ce)	
	$\mid$	$[\gamma]ce$	M
se	$::=$		surface expressions
	$\mid$	x	
	$\mid$	lit	integer literal
	$\mid$	$blit$	boolean literal
	$\mid$	$\mathsf{let}\, pat = se_1; se_2$	
	$\mid$	$(se_1, \ldots, se_n)$	
	$\mid$	$L\, se$	
	$\mid$	$\mathbf{case}\, se\, \mathbf{of}\, \{pat_1 \to se_1 \mid \ldots \mid pat_n \to se_n\}$	
	$\mid$	$\mathbf{iter}\,(pat = se_1)\{se_2\}$	
	$\mid$	$prim\, se$	
	$\mid$	$\mathbf{if}\, se_1\, \mathbf{then}\, se_2\, \mathbf{else}\, se_3$	conditional
	$\mid$	$se : \tau$	
	$\mid$	$f\, se$	
	$\mid$	$se_1 \doteq se_2$	
	$\mid$	$\mathbf{take}\, pat = se_1; se_2$	
	$\mid$	$\mathbf{return}\, se$	
	$\mid$	$\mathbf{fail}$	failure
$prog$	$::=$		surface programs
	$\mid$	$\mathsf{fun}\, f : \tau \to_{eff} \tau' = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \quad prog$	

	$\mid$	$\textbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\} prog$	
	$\mid$	$\textbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 \mid .. \mid L_n : A_n\} prog$	
	$\mid$	$\mathsf{main} : \tau_{\mathit{eff}} = se$	
$cprog$	$::=$		core programs
	$\mid$	$\mathsf{fun}\, f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce \quad cprog$	
	$\mid$	$\textbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\} cprog$	
	$\mid$	$\textbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 \mid .. \mid L_n : A_n\} cprog$	
	$\mid$	$\mathsf{main} : \tau_{\mathit{eff}} = ce$	
$rprog$	$::=$		refined programs
	$\mid$	$\textbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\} rprog$	
	$\mid$	$\textbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 \mid .. \mid L_n : A_n\} rprog$	
	$\mid$	$\mathsf{fun}\, f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce \quad rprog$	
	$\mid$	$\mathsf{fun}\, f(q) : Pf_1 \multimap Pf_2\, [\mathit{eff}] = crt \quad rprog$	
	$\mid$	$\mathsf{main} : Pf_{\mathit{eff}} = crt$	
pat	$::=$		patterns
	$\mid$	x	
	$\mid$	$L\, pat$	
	$\mid$	$(pat_1, .., pat_n)$	
β	$::=$		pattern bindings
	$\mid$	$\cdot$	
	$\mid$	$pat : \tau, \beta$	
	$\mid$	$pat_1 : \tau_1, .., pat_n : \tau_n, \beta$	
	$\mid$	$pat : \tau$	
$\mathcal{E}$	$::=$		evaluation contexts
	$\mid$	$[\cdot]$	
	$\mid$	$\mathsf{let}\, x = ce; \mathcal{E}$	
$\overline{y}$	$::=$		scrutinee variables
	$\mid$	$\cdot$	
	$\mid$	$y, \overline{y}$	
	$\mid$	$y_1, .., y_n, \overline{y}$	
M	$::=$		match branches
	$\mid$	$\cdot$	
	$\mid$	$\beta \rightarrow [\Gamma][\mathcal{E}]se \mid M$	
	$\mid$	$\beta \rightarrow [\Gamma][\mathcal{E}]se$	
$formula$	$::=$		
	$\mid$	$judgement$	
	$\mid$	$x{:}_{\mathit{eff}}\tau \in \Gamma$	
	$\mid$	$f{:}\tau \rightarrow_{\mathit{eff}} \tau' \in \Sigma_c$	
	$\mid$	$A\ \mathsf{eqtype}$	
	$\mid$	$\textbf{sort}\, D(a_1, .., a_i)\, \textbf{in}\, \Sigma_c$	
	$\mid$	$L{:}\tau \in \mathrm{ctors}(D) \in \Sigma_c$	

	$\mid$	$\textbf{sort}\, D(a_1, .., a_k) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\}\, \textbf{in}\, \Sigma_c$
	$\mid$	$\textbf{type}\, D(a_1, .., a_i)\, \textbf{in}\, \Sigma_c$
	$\mid$	$L{:}A \in \mathsf{tyctors}(D) \in \Sigma_c$
	$\mid$	$\textbf{type}\, D(a_1, .., a_k) = \{L_1 : A_1 \mid .. \mid L_n : A_n\}\, \textbf{in}\, \Sigma_c$
	$\mid$	$\mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c$
	$\mid$	$a{:}kind \in \Gamma$
	$\mid$	$kind \leq kind'$
	$\mid$	$L \neq L'$
	$\mid$	$M = \overline{pat{:}\tau, \cdot \to [\cdot][\cdot]\, se}$
	$\mid$	$ce = \mathcal{E}[ce']$
	$\mid$	$D \neq D'$
	$\mid$	$x \neq y$
	$\mid$	$a \neq b$
	$\mid$	$n = k$
	$\mid$	$\mathit{eff} = \mathit{eff}_1 \vee \mathit{eff}_2$
	$\mid$	$\mathit{eff} = \mathit{eff}_1 \vee \mathit{eff}_2 \vee \mathit{eff}_3$
	$\mid$	$\mathit{eff} = \mathit{eff}_1 \ \vee\ .. \ \vee\ \mathit{eff}_n$
	$\mid$	$\Gamma = x_1 : \tau_1[\mathit{eff}_1], .., x_n : \tau_n[\mathit{eff}_n]$
	$\mid$	$\Gamma = \overline{a : k}$
	$\mid$	$formula_1 \quad .. \quad formula_n$
	$\mid$	$x{:}ce\ [\mathsf{log}] \in \Delta$
	$\mid$	$x{:}ce@ce'\ [\mathsf{res}(u)] \in \Delta$
	$\mid$	$x{:}_{\mathit{eff}}\tau \in \Delta$
	$\mid$	$u \in \{1, ?\}$
	$\mid$	$u \in \{0, ?\}$
	$\mid$	$u = u_1 \sqcap u_2$
	$\mid$	$\Delta' = \mathsf{affinize}(\Delta)$
	$\mid$	$\mathsf{zero}(\Delta)$
	$\mid$	$\Gamma = \lvert\Delta\rvert$
	$\mid$	$\Gamma = \mathsf{Bind}(\cdot; Pf)$
	$\mid$	$\mathcal{C} = \Delta \Rightarrow \mathcal{C}'$
	$\mid$	$f(\mathbf{x} : \tau) \to_{\mathit{eff}} \tau' = ce \in \Sigma_c$
	$\mid$	$\mathit{eff}' \in \{\mathbf{S}, \lfloor \mathit{eff} \rfloor\}$
	$\mid$	$f{:}RF \in \Sigma_c$
	$\mid$	$f{:}RF \in \Sigma_r$
	$\mid$	$f{:}\tau \to_{\mathit{eff}} \tau' \in \Sigma_r$
	$\mid$	$f(\mathbf{x} : \tau) \to_{\mathit{eff}} \tau' = ce \in \Sigma_r$
	$\mid$	$\mathsf{len}(\Delta) = \mathsf{len}(\Delta')$
	$\mid$	$\Delta = \Delta'$
	$\mid$	$A = \mathsf{Prod}(\overline{A})$
	$\mid$	$prim{:}RF$
	$\mid$	$ce = \mathsf{branch}(L, x, ce')$
u	$::=$	usage flags
	$\mid$	0
	$\mid$	1
	$\mid$	$?$
Δ	$::=$	refined contexts
	$\mid$	$\cdot$

	$\mid$	$x_1 : \tau_1[\mathit{eff}_1],\ ..,\ x_n : \tau_n[\mathit{eff}_n]$	
	$\mid$	$x_1 : ce_1[\,\mathsf{log}\,],\ ..,\ x_n : ce_n[\,\mathsf{log}\,]$	
	$\mid$	$x_1 : ce_1@ce_1'[\,\mathsf{res}\,(u_1)],\ ..,\ x_n : ce_n@ce_n'[\,\mathsf{res}\,(u_n)]$	
	$\mid$	Δ, Δ'	
$\overline{\Delta}$	$::=$		context lists
	$\mid$	$\Delta; \overline{\Delta}$	
	$\mid$	$\cdot$	
	$\mid$	$(\overline{\Delta})$	
	$\mid$	$\Delta_1;\ ..;\Delta_n$	
$\mathcal{C}$	$::=$		SMT constraints
	$\mid$	$\top$	
	$\mid$	$\mathcal{C}_1\ \wedge\ \mathcal{C}_2$	
	$\mid$	$\forall\, x_1 : \tau_1\, ..\, \forall\, x_n : \tau_n.\mathcal{C}$	
	$\mid$	$ce\ \Rightarrow\ \mathcal{C}$	
	$\mid$	ce	
	$\mid$	$\bot$	
	$\mid$	$(\mathcal{C})$	
	$\mid$	$\mathsf{is}\, ce\, L$	
Pf	$::=$		proof sorts
	$\mid$	$\cdot$	
	$\mid$	$x{:}_{\mathit{eff}}\tau, Pf$	
	$\mid$	$ce\ [\mathsf{log}], Pf$	
	$\mid$	$ce@ce'\ [\mathsf{res}], Pf$	
	$\mid$	$(y).ce\ [\mathsf{res}], Pf$	
	$\mid$	$[\gamma]Pf$	M
RF	$::=$		refined function types
	$\mid$	$Pf_1 \multimap Pf_2\ [\mathit{eff}]$	
$cpat$	$::=$		core patterns
	$\mid$	x	
	$\mid$	$(cpat_1,\ ..,\ cpat_n)$	
$lpat$	$::=$		logical patterns
	$\mid$	x	
	$\mid$	auto	
$rpat$	$::=$		resource patterns
	$\mid$	x	
	$\mid$	$\mathsf{return}\ lpat$	
	$\mid$	$\mathsf{take}(cpat, rpat_1); rpat_2$	
	$\mid$	fail	
	$\mid$	$\mathsf{let}\mathbf{w}]\ cpat; rpat$	
	$\mid$	$\mathsf{case}\ L(cpat); lpat; rpat$	
	$\mid$	$\mathsf{iftrue}; rpat$	
	$\mid$	$\mathsf{iffalse}; rpat$	

	|	$\mathsf{unfold}; rpat$	
	|	$\mathsf{annot}; rpat$	
q	::=		refined patterns
	|	$\cdot$	
	|	$(cpat, q)$	
	|	$(\mathsf{log}\ lpat, q)$	
	|	$(\mathsf{res}\ rpat, q)$	
	|	$(\mathsf{do}\ cpat = rpat, q)$	
crt	::=		core refined terms
	|	$\mathsf{let}\ q = crt_1; crt_2$	
	|	$\mathsf{let}\ \mathsf{log}\ x = lpf; crt$	
	|	$\mathsf{let}\ \mathsf{res}\ x = rpf; crt$	
	|	$\mathsf{let}\ \mathsf{core}\ [w]x = ce; crt$	
	|	$\mathsf{let}\ \mathsf{core}\ [w](x_1, \ldots, x_n) = ce; crt$	
	|	$crt : Pf$	
	|	$prim\ rsp$	
	|	$f\ rsp$	
	|	(rsp)	
	|	$\mathbf{iter}\ [ce](q = crt_1)\{crt_2\}$	
	|	$\mathbf{if}\ [x]ce\ \mathbf{then}\ crt_1\ \mathbf{else}\ crt_2$	
	|	$\mathbf{case}\ [y]ce\ \mathbf{of}\ \{L_1\ x_1 \to crt_1 \mid \ldots \mid L_n\ x_n \to crt_n\}$	
	|	$\mathsf{exfalso}$	
	|	$\mathsf{open\text{-}take}\ rpf$	
rsp	::=		refined spines
	|	$\cdot$	
	|	ce, rsp	
	|	rpf, rsp	
	|	lpf, rsp	
lpf	::=		logical proof facts
	|	x	
	|	auto	
	|	$\mathsf{unfold}\ f(ce)$	
	|	$\mathsf{open\text{-}ret}\ rpf$	
	|	$lpf : ce$	
rpf	::=		resource proof facts
	|	x	
	|	$\mathsf{make\text{-}ret}\ lpf$	
	|	$\mathsf{make\text{-}take}\ crt$	
	|	$rpf : ce@ce'$	
$Jtype$	::=		
	|	$kind \leq kind'$	
	|	$\Sigma_c; \Gamma \vdash \tau{:}kind$	
	|	$\Sigma_c; \Gamma; D'(\overline{a}) \vdash,\ \mathsf{guarded}$	

	$\mid$	$\Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok}$
	$\mid$	$\Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok}$
	$\mid$	$\Sigma_c; \Gamma \vdash \gamma : \Gamma'$
	$\mid$	$\gamma = \mathsf{id}(\Gamma)$
	$\mid$	$\tau' = [\gamma]\tau$
	$\mid$	$ce' = [\gamma]ce$
	$\mid$	$Pf' = [\gamma]Pf$
	$\mid$	$\gamma'' = \gamma\,;\gamma'$
	$\mid$	$A\ \mathsf{eqtype}$
	$\mid$	$\mathit{eff} \leq \mathit{eff}'$
	$\mid$	$\mathit{eff}' = \lfloor \mathit{eff} \rfloor$
	$\mid$	$\mathit{eff} = \mathit{eff}_1 \vee \mathit{eff}_2$
	$\mid$	$\mathit{prim}{:}\tau \rightarrow_{\mathit{eff}} \tau'$
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Rightarrow \tau$
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Leftarrow \tau$
	$\mid$	$\Sigma_c \vdash \mathit{cprog}$
Jcore	$::=$	
	$\mid$	$L : \tau\ \mathbf{in}\ D(\tau_1, \ldots, \tau_i)\ \mathbf{in}\ \Sigma_c$
	$\mid$	$D(a_1, \ldots, a_k) = \{L_1 : \tau_1 \mid \ldots \mid L_n : \tau_n\} \in \Sigma_c$
$\mathit{Jelaborate}$	$::=$	
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Rightarrow \tau \rightsquigarrow ce$
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau \rightsquigarrow ce$
	$\mid$	$\Sigma_s \vdash \mathit{prog}$ $\rightsquigarrow \Sigma_c \vdash \mathit{cprog}$
Jmatch	$::=$	
	$\mid$	$M\ \mathsf{has_con}$
	$\mid$	$M\ \mathsf{has_tup}$
	$\mid$	$M/\mathit{var} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'$
	$\mid$	$M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'$
	$\mid$	$M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'$
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \overline{y}; M \Leftarrow_{\epsilon} \tau \rightsquigarrow ce$
	$\mid$	$\mathcal{E}[ce] \rightsquigarrow ce'$
$\mathit{Jrefine}$	$::=$	
	$\mid$	$\Sigma_r \vdash \Delta\ \mathsf{wf}$
	$\mid$	$\Gamma = \lvert\Delta\rvert$
	$\mid$	$\Delta' = \mathsf{affinize}(\Delta)$
	$\mid$	$\mathsf{zero}(\Delta)$
	$\mid$	$\mathcal{C} = \Delta \Rightarrow \mathcal{C}'$
	$\mid$	$\Sigma_r; \Gamma \vdash \mathcal{C}\ \mathsf{wf}$
	$\mid$	$\Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf\ \mathsf{wf}$
	$\mid$	$\Sigma_r \vdash RF\ \mathsf{wf}$
	$\mid$	$\Sigma_r \vdash f{:}RF$
	$\mid$	$\overline{A} = \{Pf\}$
	$\mid$	$\Sigma_c = \mathsf{Comp}(\Sigma_r)$
	$\mid$	$\vdash \Sigma_r\ \mathsf{wf}$
	$\mid$	$\Sigma_r; \Delta \vdash \mathit{lpf} \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}$
	$\mid$	$\Sigma_r; \Delta \vdash \mathit{lpf} \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}$

	|	$\Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'$
	|	$\Sigma_r; \Delta \vdash rpf \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}$
	|	$\Sigma_r; \Delta \vdash rpf \Leftarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}$
	|	$\Sigma_r; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C}$
	|	$\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta$
	|	$\Sigma_r; \Delta \vdash_{\mathit{eff}} q : Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
	|	$\Sigma_r; \Delta \vdash_{\mathit{eff}} crt \Rightarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
	|	$\Sigma_r; \Delta \vdash_{\mathit{eff}} y{:}ce \ \mathsf{is}\ L\,x \rightarrow crt \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
	|	$\Sigma_r; \Delta \vdash_{\mathit{eff}} crt \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
	|	$\Sigma_r; \Delta \vdash_{\mathit{eff}} rsp : Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
	|	$\Sigma_r; \Delta \vdash_{\mathit{eff}} rsp : Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
	|	$\Sigma_r \vdash rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C}$
	|	$\Sigma_r \vdash \sqcap\overline{\Delta} = \Delta$
judgement	::=	
	|	*Jtype*
	|	*Jcore*
	|	*Jelaborate*
	|	*Jmatch*
	|	*Jrefine*
user_syntax	::=	
	|	*var*
	|	*label*
	|	*lit*
	|	*f*
	|	*dsort*
	|	*tvar*
	|	*index*
	|	*terminals*
	|	*blit*
	|	*eff*
	|	*kind*
	|	τ
	|	$\overline{A}$
	|	F
	|	Γ
	|	γ
	|	Σ_c
	|	Σ_s
	|	Σ_r
	|	*prim*
	|	*ce*
	|	*se*
	|	*prog*
	|	*cprog*
	|	*rprog*
	|	*pat*
	|	β
	|	$\mathcal{E}$

$$
\begin{array}{|l}
\overline{y} \\
M \\
formula \\
u \\
\Delta \\
\overline{\Delta} \\
\mathcal{C} \\
Pf \\
RF \\
cpat \\
lpat \\
rpat \\
q \\
crt \\
rsp \\
lpf \\
rpf
\end{array}
$$

$\boxed{kind \leq kind'}$

$$\frac{}{\mathsf{type} \leq \mathsf{type}} \quad \text{SK_TYPE_TYPE}$$

$$\frac{}{\mathsf{type} \leq \mathsf{sort}} \quad \text{SK_TYPE_SORT}$$

$$\frac{}{\mathsf{sort} \leq \mathsf{sort}} \quad \text{SK_SORT_SORT}$$

$\boxed{\Sigma_c; \Gamma \vdash \tau{:}kind}$

$$\frac{}{\Sigma_c; \Gamma \vdash \mathsf{Int}{:}kind} \quad \text{KWF_INT}$$

$$\frac{}{\Sigma_c; \Gamma \vdash \mathsf{Bool}{:}kind} \quad \text{KWF_BOOL}$$

$$\frac{\Sigma_c; \Gamma \vdash \tau{:}\mathsf{type}}{\Sigma_c; \Gamma \vdash \mathsf{Ptr}\ \tau{:}kind} \quad \text{KWF_PTR}$$

$$\frac{\Sigma_c; \Gamma \vdash \tau_1{:}kind \quad .. \quad \Sigma_c; \Gamma \vdash \tau_n{:}kind}{\Sigma_c; \Gamma \vdash (\tau_1, \ .., \tau_n){:}kind} \quad \text{KWF_RECORD}$$

$$\frac{\begin{array}{c}\textbf{type}\ D(a_1, \ .., a_k)\ \textbf{in}\ \Sigma_c \quad n = k \\ \Sigma_c; \Gamma \vdash \tau_1{:}\mathsf{type} \quad .. \quad \Sigma_c; \Gamma \vdash \tau_n{:}\mathsf{type}\end{array}}{\Sigma_c; \Gamma \vdash D(\tau_1, \ .., \tau_n){:}\mathsf{type}} \quad \text{KWF_TYPE_APP}$$

$$\frac{\begin{array}{c}\textbf{sort}\ D(a_1, \ .., a_k)\ \textbf{in}\ \Sigma_c \quad n = k \\ \Sigma_c; \Gamma \vdash \tau_1{:}\mathsf{sort} \quad .. \quad \Sigma_c; \Gamma \vdash \tau_n{:}\mathsf{sort}\end{array}}{\Sigma_c; \Gamma \vdash D(\tau_1, \ .., \tau_n){:}\mathsf{sort}} \quad \text{KWF_SORT_APP}$$

$$\frac{\Sigma_c; \Gamma \vdash \tau{:}\mathsf{sort}}{\Sigma_c; \Gamma \vdash \mathsf{Pred}\ \tau{:}\mathsf{sort}} \quad \text{KWF_PRED}$$

$$\frac{a{:}kind' \in \Gamma \quad kind' \leq kind}{\Sigma_c; \Gamma \vdash a{:}kind} \quad \text{KWF_TVAR}$$

$\boxed{\Sigma_c; \Gamma; D'(\overline{a}) \vdash, \ \mathsf{guarded}}$

$$\frac{}{\Sigma_c; \Gamma; D'(\overline{a}) \vdash, \ \mathsf{guarded}} \quad \textsc{TWfG_int}$$

$$\frac{}{\Sigma_c; \Gamma; D'(\overline{a}) \vdash, \ \mathsf{guarded}} \quad \textsc{TWfG_bool}$$

$$\frac{\Sigma_c, \mathsf{type}\ D'(\overline{a}) = \{\overline{L : A}\}; \Gamma \vdash \tau{:}\mathsf{type}}{\Sigma_c; \Gamma; D'(\overline{a}) \vdash, \ \mathsf{guarded}} \quad \textsc{TWfG_ptr}$$

$$\frac{\Sigma_c; \Gamma; D'(\overline{a}) \vdash, \ \mathsf{guarded} \quad .. \quad \Sigma_c; \Gamma; D'(\overline{a}) \vdash, \ \mathsf{guarded}}{\Sigma_c; \Gamma; D'(\overline{a}) \vdash, \ \mathsf{guarded}} \quad \textsc{TWfG_record}$$

$$\frac{\begin{array}{c}\mathbf{type}\, D(b_1, .., b_k)\, \mathbf{in}\, \Sigma_c \quad n = k \quad D \neq D' \\ \Sigma_c; \Gamma \vdash \tau_1{:}\mathsf{type} \quad .. \quad \Sigma_c; \Gamma \vdash \tau_n{:}\mathsf{type}\end{array}}{\Sigma_c; \Gamma; D'(\overline{a}) \vdash, \ \mathsf{guarded}} \quad \textsc{TWfG_app}$$

$$\frac{a{:}kind \in \Gamma}{\Sigma_c; \Gamma; D'(\overline{a}) \vdash, \ \mathsf{guarded}} \quad \textsc{TWfG_tvar}$$

$\boxed{\Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok}}$

$$\frac{\begin{array}{c}\Gamma = \overline{a : k} \\ \Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}; \Gamma \vdash \tau_1{:}\mathsf{sort} \quad .. \quad \Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}; \Gamma \vdash \tau_n{:}\mathsf{sort}\end{array}}{\Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok}} \quad \textsc{SDWf_ok}$$

$\boxed{\Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok}}$

$$\frac{\begin{array}{c}\Gamma = \overline{a : k} \\ \Sigma_c; \Gamma; D(\overline{a}) \vdash, \ \mathsf{guarded} \quad .. \quad \Sigma_c; \Gamma; D(\overline{a}) \vdash, \ \mathsf{guarded}\end{array}}{\Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok}} \quad \textsc{TDWf_ok}$$

$\boxed{\Sigma_c; \Gamma \vdash \gamma : \Gamma'}$

$$\frac{}{\Sigma_c; \Gamma \vdash \cdot : \cdot} \quad \textsc{SubWf_empty}$$

$$\frac{\Sigma_c; \Gamma \vdash \gamma : \Gamma' \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}}\ ce \Rightarrow \tau}{\Sigma_c; \Gamma \vdash \gamma, ce/x : \Gamma', x{:}_{\mathit{eff}}\tau} \quad \textsc{SubWf_term}$$

$$\frac{\Sigma_c; \Gamma \vdash \gamma : \Gamma' \quad \Sigma_c; \Gamma \vdash \tau{:}kind}{\Sigma_c; \Gamma \vdash \gamma, \tau/a : \Gamma', a{:}kind} \quad \textsc{SubWf_type}$$

$\boxed{\gamma = \mathsf{id}(\Gamma)}$

$$\frac{}{\cdot = \mathsf{id}(\cdot)} \quad \textsc{IdDef_empty}$$

$$\frac{\gamma = \mathsf{id}(\Gamma)}{\gamma, x/x = \mathsf{id}(\Gamma, x{:}_{\mathit{eff}}\tau)} \quad \textsc{IdDef_term}$$

$$\frac{\gamma = \mathsf{id}(\Gamma)}{\gamma, a/a = \mathsf{id}(\Gamma, a{:}kind)} \quad \textsc{IdDef_type}$$

$\boxed{\tau' = [\gamma]\tau}$

$$\frac{}{\mathsf{Int} = [\gamma]\mathsf{Int}} \quad \textsc{SS_int}$$

$$\frac{}{\mathsf{Bool} = [\gamma]\mathsf{Bool}} \quad \textsc{SS_bool}$$

$$\frac{\tau' = [\gamma]\tau}{\mathsf{Ptr}\ \tau' = [\gamma]\mathsf{Ptr}\ \tau} \quad \text{SS_PTR}$$

$$\frac{\tau_1' = [\gamma]\tau_1 \quad .. \quad \tau_n' = [\gamma]\tau_n}{(\tau_1', .., \tau_n') = [\gamma](\tau_1, .., \tau_n)} \quad \text{SS_RECORD}$$

$$\frac{\tau_1' = [\gamma]\tau_1 \quad .. \quad \tau_n' = [\gamma]\tau_n}{D(\tau_1', .., \tau_n') = [\gamma]D(\tau_1, .., \tau_n)} \quad \text{SS_APP}$$

$$\frac{\tau' = [\gamma]\tau}{\mathsf{Pred}\ \tau' = [\gamma]\mathsf{Pred}\ \tau} \quad \text{SS_PRED}$$

$$\frac{}{\tau = [\gamma, \tau/a]a} \quad \text{SS_TVAR_HERE}$$

$$\frac{a \neq b \quad \tau' = [\gamma]b}{\tau' = [\gamma, \tau/a]b} \quad \text{SS_TVAR_SKIP_TY}$$

$$\frac{\tau' = [\gamma]b}{\tau' = [\gamma, ce/x]b} \quad \text{SS_TVAR_SKIP_TM}$$

$\boxed{ce' = [\gamma]ce}$

$$\frac{}{ce = [\gamma, ce/x]x} \quad \text{TS_VAR_HERE}$$

$$\frac{x \neq y \quad ce' = [\gamma]x}{ce' = [\gamma, ce/y]x} \quad \text{TS_VAR_SKIP_TM}$$

$$\frac{ce' = [\gamma]x}{ce' = [\gamma, \tau/a]x} \quad \text{TS_VAR_SKIP_TY}$$

$$\frac{}{lit = [\gamma]lit} \quad \text{TS_INTLIT}$$

$$\frac{}{blit = [\gamma]blit} \quad \text{TS_BOOLLIT}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad ce_2' = [\gamma, x/x]ce_2}{\mathsf{let}\ x = ce_1'; ce_2' = [\gamma]\mathsf{let}\ x = ce_1; ce_2} \quad \text{TS_LET}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad .. \quad ce_n' = [\gamma]ce_n}{(ce_1', .., ce_n') = [\gamma](ce_1, .., ce_n)} \quad \text{TS_TUPLE}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad ce_2' = [\gamma, x_1/x_1, .., x_n/x_n]ce_2}{\mathsf{let}\ (x_1, .., x_n) = ce_1'; ce_2' = [\gamma]\mathsf{let}\ (x_1, .., x_n) = ce_1; ce_2} \quad \text{TS_LETTUPLE}$$

$$\frac{ce' = [\gamma]ce}{L\ ce' = [\gamma]L\ ce} \quad \text{TS_INJECT}$$

$$\frac{ce' = [\gamma]ce \quad ce_1' = [\gamma, x_1/x_1]ce_1 \quad .. \quad ce_n' = [\gamma, x_n/x_n]ce_n}{\textbf{case}\ ce'\ \textbf{of}\ \{L_1\ x_1 \to ce_1' | .. | L_n\ x_n \to ce_n'\} = [\gamma]\textbf{case}\ ce\ \textbf{of}\ \{L_1\ x_1 \to ce_1 | .. | L_n\ x_n \to ce_n\}} \quad \text{TS_CASE}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad ce_2' = [\gamma, x/x]ce_2}{\textbf{iter}\ (x = ce_1')\{ce_2'\} = [\gamma]\textbf{iter}\ (x = ce_1)\{ce_2\}} \quad \text{TS_ITER}$$

$$\frac{ce' = [\gamma]ce}{prim\ ce' = [\gamma]prim\ ce} \quad \text{TS_PRIM}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad ce_2' = [\gamma]ce_2 \quad ce_3' = [\gamma]ce_3}{\textbf{if}\ ce_1'\ \textbf{then}\ ce_2'\ \textbf{else}\ ce_3' = [\gamma]\textbf{if}\ ce_1\ \textbf{then}\ ce_2\ \textbf{else}\ ce_3} \quad \text{TS_IF}$$

$$\frac{ce' = [\gamma]ce \quad \tau' = [\gamma]\tau}{ce' : \tau' = [\gamma]ce : \tau} \quad \text{TS_ANNOT}$$

$$\frac{ce' = [\gamma]ce}{f\ ce' = [\gamma]f\ ce} \quad \text{TS_CALL}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad ce_2' = [\gamma]ce_2}{ce_1' \doteq ce_2' = [\gamma]ce_1 \doteq ce_2} \quad \text{TS_EQ}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad ce_2' = [\gamma, x/x]ce_2}{\textbf{take}\, x = ce_1'; ce_2' = [\gamma]\textbf{take}\, x = ce_1; ce_2} \quad \text{TS_TAKE}$$

$$\frac{ce' = [\gamma]ce}{\textbf{return}\, ce' = [\gamma]\textbf{return}\, ce} \quad \text{TS_RETURN}$$

$\boxed{Pf' = [\gamma]Pf}$

$$\frac{}{\cdot = [\gamma]\cdot} \quad \text{PS_NIL}$$

$$\frac{\tau' = [\gamma]\tau \quad Pf' = [\gamma, x/x]Pf}{x{:}_{\mathit{eff}}\tau', Pf' = [\gamma]x{:}_{\mathit{eff}}\tau, Pf} \quad \text{PS_COMP}$$

$$\frac{ce' = [\gamma]ce \quad Pf' = [\gamma]Pf}{ce'\ [\mathsf{log}], Pf' = [\gamma]ce\ [\mathsf{log}], Pf} \quad \text{PS_LOG}$$

$$\frac{ce_1' = [\gamma]ce_1 \quad ce_2' = [\gamma]ce_2 \quad Pf' = [\gamma]Pf}{ce_1'@ce_2'\ [\mathsf{res}], Pf' = [\gamma]ce_1@ce_2\ [\mathsf{res}], Pf} \quad \text{PS_RES}$$

$$\frac{ce' = [\gamma, y/y]ce \quad Pf' = [\gamma]Pf}{(y).ce'\ [\mathsf{res}], Pf' = [\gamma](y).ce\ [\mathsf{res}], Pf} \quad \text{PS_DEPRES}$$

$\boxed{\gamma'' = \gamma\,;\gamma'}$

$$\frac{}{\cdot = \gamma\,;\cdot} \quad \text{CompDef_empty}$$

$$\frac{\gamma'' = \gamma\,;\gamma' \quad ce' = [\gamma]ce}{\gamma'', ce'/x = \gamma\,;\gamma', ce/x} \quad \text{CompDef_term}$$

$$\frac{\gamma'' = \gamma\,;\gamma' \quad \tau' = [\gamma]\tau}{\gamma'', \tau'/a = \gamma\,;\gamma', \tau/a} \quad \text{CompDef_type}$$

$\boxed{A\ \mathsf{eqtype}}$

$$\frac{}{\mathsf{Int\ eqtype}} \quad \text{EqTy_int}$$

$$\frac{}{\mathsf{Bool\ eqtype}} \quad \text{EqTy_bool}$$

$$\frac{}{\mathsf{Ptr}\ A\ \mathsf{eqtype}} \quad \text{EqTy_ptr}$$

$\boxed{\mathit{eff} \leq \mathit{eff}'}$

$$\frac{}{\mathbf{0} \leq \mathbf{0}} \quad \text{Sub_pure_pure}$$

$$\frac{}{\mathbf{0} \leq \mathbf{1}} \quad \text{Sub_pure_impure}$$

$$\frac{}{\mathbf{1} \leq \mathbf{1}} \quad \textsc{Sub_impure_impure}$$

$$\frac{}{\mathbf{0} \leq \mathbf{S}} \quad \textsc{Sub_pure_spec}$$

$$\frac{}{\mathbf{S} \leq \mathbf{S}} \quad \textsc{Sub_spec_spec}$$

$\boxed{\mathit{eff}' = \lfloor \mathit{eff} \rfloor}$

$$\frac{}{\mathbf{0} = \lfloor \mathbf{0} \rfloor} \quad \textsc{Pur_pure}$$

$$\frac{}{\mathbf{0} = \lfloor \mathbf{1} \rfloor} \quad \textsc{Pur_impure}$$

$$\frac{}{\mathbf{S} = \lfloor \mathbf{S} \rfloor} \quad \textsc{Pur_spec}$$

$\boxed{\mathit{eff} = \mathit{eff}_1 \vee \mathit{eff}_2}$

$$\frac{}{\mathit{eff} = \mathit{eff} \vee \mathit{eff}} \quad \textsc{Join_refl}$$

$$\frac{}{\mathbf{1} = \mathbf{0} \vee \mathbf{1}} \quad \textsc{Join_pure_impure}$$

$$\frac{}{\mathbf{1} = \mathbf{1} \vee \mathbf{0}} \quad \textsc{Join_impure_pure}$$

$$\frac{}{\mathbf{S} = \mathbf{0} \vee \mathbf{S}} \quad \textsc{Join_pure_spec}$$

$$\frac{}{\mathbf{S} = \mathbf{S} \vee \mathbf{0}} \quad \textsc{Join_spec_pure}$$

$\boxed{\mathit{prim}{:}\tau \rightarrow_{\mathit{eff}} \tau'}$

$$\frac{}{\mathbf{Add}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{0}} \mathsf{Int}} \quad \textsc{PT_add}$$

$$\frac{}{\mathbf{Sub}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{0}} \mathsf{Int}} \quad \textsc{PT_sub}$$

$$\frac{}{\mathbf{Mul}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{0}} \mathsf{Int}} \quad \textsc{PT_mul}$$

$$\frac{}{\mathbf{Div}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{1}} \mathsf{Int}} \quad \textsc{PT_div}$$

$$\frac{}{\mathbf{And}{:}(\mathsf{Bool}, \mathsf{Bool}) \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \textsc{PT_and}$$

$$\frac{}{\mathbf{Or}{:}(\mathsf{Bool}, \mathsf{Bool}) \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \textsc{PT_or}$$

$$\frac{}{\mathbf{Not}{:}\mathsf{Bool} \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \textsc{PT_not}$$

$$\frac{}{\mathbf{Lt}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \textsc{PT_lt}$$

$$\frac{}{\mathbf{Le}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \textsc{PT_le}$$

$$\frac{}{\mathbf{Gt}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \textsc{PT_gt}$$

$$\frac{}{\mathbf{Ge}{:}(\mathsf{Int}, \mathsf{Int}) \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \textsc{PT_ge}$$

$$\frac{A \ \mathsf{eqtype}}{\mathbf{Eq}\,[A]{:}(A, A) \rightarrow_{\mathbf{0}} \mathsf{Bool}} \quad \textsc{PT_eq}$$

$$\frac{}{\textbf{New}\,[A]{:}A \rightarrow_{\mathbf{1}} \mathsf{Ptr}\ A}\ \text{PT_NEW}$$

$$\frac{}{\textbf{Del}\,[A]{:}\mathsf{Ptr}\ A \rightarrow_{\mathbf{1}} (\,)}\ \text{PT_DEL}$$

$$\frac{}{\textbf{Get}\,[A]{:}\mathsf{Ptr}\ A \rightarrow_{\mathbf{1}} A}\ \text{PT_GET}$$

$$\frac{}{\textbf{Set}\,[A]{:}(\mathsf{Ptr}\ A, A) \rightarrow_{\mathbf{1}} (\,)}\ \text{PT_SET}$$

$$\frac{}{\textbf{Own}\,[A]{:}\mathsf{Ptr}\ A \rightarrow_{\mathbf{S}} \mathsf{Pred}\ (A)}\ \text{PT_OWN}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Rightarrow \tau}$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{lit} \Rightarrow \mathsf{Int}}\ \text{SYN_INT}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{blit} \Rightarrow \mathsf{Bool}}\ \text{SYN_BOOL}$$

$$\frac{x{:}_{\mathit{eff}'}\tau \in \Gamma \quad \mathit{eff}' \leq \mathit{eff}_0}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x \Rightarrow \tau}\ \text{SYN_VAR}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Leftarrow \tau}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce : \tau \Rightarrow \tau}\ \text{SYN_ANNOT}$$

$$\frac{\begin{array}{l} \mathit{prim}{:}\tau \rightarrow_{\mathit{eff}'} \tau' \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} ce \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{prim}\ ce \Rightarrow \tau'}\ \text{SYN_APP}$$

$$\frac{\begin{array}{l} f{:}\tau \rightarrow_{\mathit{eff}'} \tau' \in \Sigma_c \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} ce \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} f\ ce \Rightarrow \tau'}\ \text{SYN_CALL}$$

$$\frac{\begin{array}{l} A\ \mathsf{eqtype} \\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce_1 \Rightarrow A \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce_2 \Leftarrow A \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce_1 \doteq ce_2 \Rightarrow \mathsf{Bool}}\ \text{SYN_EQ}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Leftarrow \tau}$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce_1 \Rightarrow \tau \\ \epsilon = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma, x{:}_{\epsilon}\tau \vdash_{\mathit{eff}_0} ce_2 \Leftarrow \tau' \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathsf{let}\ x = ce_1; ce_2 \Leftarrow \tau'}\ \text{CHK_LET}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce_1 \Leftarrow \tau_1 \quad .. \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce_n \Leftarrow \tau_n}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} (ce_1, .., ce_n) \Leftarrow (\tau_1, .., \tau_n)}\ \text{CHK_TUPLE}$$

$$\frac{\begin{array}{l} \epsilon = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\epsilon} ce_1 \Rightarrow (\tau_1, .., \tau_n) \\ \Gamma' = x_1 : \tau_1[\epsilon], .., x_n : \tau_n[\epsilon] \\ \Sigma_c; \Gamma, \Gamma' \vdash_{\mathit{eff}_0} ce_2 \Leftarrow \tau' \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathsf{let}\ (x_1, .., x_n) = ce_1; ce_2 \Leftarrow \tau'}\ \text{CHK_LETTUPLE}$$

$$\frac{\begin{array}{l} L : \tau \,\mathbf{in}\, D(\tau_1, \,..\,, \tau_i) \,\mathbf{in}\, \Sigma_c \\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} L\, ce \Leftarrow D(\tau_1, \,..\,, \tau_i)} \quad \textsc{Chk_inject}$$

$$\frac{\begin{array}{l} \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce \Rightarrow D(\tau_1, \,..\,, \tau_i) \\ D(a_1, \,..\,, a_k) = \{L_1 : \tau_1' | \,..\, | L_n : \tau_n'\} \in \Sigma_c \\ L_1 : \tau_1'' \,\mathbf{in}\, D(\tau_1, \,..\,, \tau_i) \,\mathbf{in}\, \Sigma_c \quad .. \quad L_n : \tau_n'' \,\mathbf{in}\, D(\tau_1, \,..\,, \tau_i) \,\mathbf{in}\, \Sigma_c \\ \Sigma_c; \Gamma, x_1{:}_{\mathit{eff}'}\tau_1'' \vdash_{\mathit{eff}_0} ce_1 \Leftarrow \tau \quad .. \quad \Sigma_c; \Gamma, x_n{:}_{\mathit{eff}'}\tau_n'' \vdash_{\mathit{eff}_0} ce_n \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{case}\, ce\, \mathbf{of}\, \{L_1\, x_1 \to ce_1 | \,..\, | L_n\, x_n \to ce_n\} \Leftarrow \tau} \quad \textsc{Chk_case}$$

$$\frac{\begin{array}{l} \mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c \\ \Sigma_c; \Gamma \vdash_{\mathbf{0}} ce_1 \Rightarrow A \\ \Sigma_c; \Gamma, x{:}_{\mathbf{0}}A \vdash_{\mathbf{1}} ce_2 \Leftarrow D(A, B) \end{array}}{\Sigma_c; \Gamma \vdash_{\mathbf{1}} \mathbf{iter}\, (x = ce_1)\{ce_2\} \Leftarrow B} \quad \textsc{Chk_iter}$$

$$\frac{\begin{array}{l} \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce_1 \Leftarrow \mathsf{Bool} \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce_2 \Leftarrow \tau \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce_3 \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{if}\, ce_1\, \mathbf{then}\, ce_2\, \mathbf{else}\, ce_3 \Leftarrow \tau} \quad \textsc{Chk_if}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \tau}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathbf{return}\, ce \Leftarrow \mathsf{Pred}\ \tau} \quad \textsc{Chk_return}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathbf{fail} \Leftarrow \mathsf{Pred}\ \tau} \quad \textsc{Chk_fail}$$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce_1 \Rightarrow \mathsf{Pred}\ \tau \\ \Sigma_c; \Gamma, x{:}_{\mathbf{S}}\tau \vdash_{\mathbf{S}} ce_2 \Leftarrow \mathsf{Pred}\ \tau' \end{array}}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathbf{take}\, x = ce_1; ce_2 \Leftarrow \mathsf{Pred}\ \tau'} \quad \textsc{Chk_take}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Rightarrow \tau}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Leftarrow \tau} \quad \textsc{Chk_sub}$$

$\boxed{\Sigma_c \vdash \mathit{cprog}}$

$$\frac{\Sigma_c; \cdot \vdash_{\mathit{eff}} ce \Leftarrow \tau}{\Sigma_c \vdash \mathsf{main} : \tau_{\mathit{eff}} = ce} \quad \textsc{CProg_main}$$

$$\frac{\begin{array}{l} \Sigma_c; \cdot, x{:}_{\mathbf{0}}A \vdash_{\mathbf{0}} ce \Leftarrow B \\ \Sigma_c, \mathsf{fun}\ f(x : A) \to_{\mathbf{0}} B = ce \vdash \mathit{cprog} \end{array}}{\Sigma_c \vdash \mathsf{fun}\ f(x : A) \to_{\mathbf{0}} B = ce \quad \mathit{cprog}} \quad \textsc{CProg_purefun}$$

$$\frac{\begin{array}{l} \Sigma_c, f : A \to_{\mathbf{1}} B; \cdot, x{:}_{\mathbf{0}}A \vdash_{\mathbf{1}} ce \Leftarrow B \\ \Sigma_c, f : A \to_{\mathbf{1}} B \vdash \mathit{cprog} \end{array}}{\Sigma_c \vdash \mathsf{fun}\ f(x : A) \to_{\mathbf{1}} B = ce \quad \mathit{cprog}} \quad \textsc{CProg_impurefun}$$

$$\frac{\begin{array}{l} \Sigma_c, f : \tau \to_{\mathbf{S}} \tau'; \cdot, x{:}_{\mathbf{S}}\tau \vdash_{\mathbf{S}} ce \Leftarrow \tau' \\ \Sigma_c, \mathsf{fun}\ f(x : \tau) \to_{\mathbf{S}} \tau' = ce \vdash \mathit{cprog} \end{array}}{\Sigma_c \vdash \mathsf{fun}\ f(x : \tau) \to_{\mathbf{S}} \tau' = ce \quad \mathit{cprog}} \quad \textsc{CProg_specfun}$$

$$\frac{\begin{array}{l} \Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok} \\ \Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} \vdash \mathit{cprog} \end{array}}{\Sigma_c \vdash \mathbf{sort}\, D(a_1, \,..\,, a_i) = \{L_1 : \tau_1 | \,..\, | L_n : \tau_n\} \mathit{cprog}} \quad \textsc{CProg_sortdecl}$$

$$\frac{\begin{array}{c}\Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok} \\ \Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} \vdash \mathit{cprog}\end{array}}{\Sigma_c \vdash \mathbf{type}\, D(a_1, \ldots, a_i) = \{L_1 : A_1 | \ldots | L_n : A_n\} \mathit{cprog}} \quad \textsc{CProg_typedecl}$$

$\boxed{L : \tau\, \mathbf{in}\, D(\tau_1, \ldots, \tau_i)\, \mathbf{in}\, \Sigma_c}$

$$\frac{\mathbf{sort}\, D(a_1, \ldots, a_i)\, \mathbf{in}\, \Sigma_c \quad L{:}\tau' \in \mathrm{ctors}(D) \in \Sigma_c \quad \tau = [\cdot, \tau_1/a_1, \ldots, \tau_i/a_i]\tau'}{L : \tau\, \mathbf{in}\, D(\tau_1, \ldots, \tau_i)\, \mathbf{in}\, \Sigma_c} \quad \textsc{Ctor_sort}$$

$$\frac{\mathbf{type}\, D(a_1, \ldots, a_i)\, \mathbf{in}\, \Sigma_c \quad L{:}A' \in \mathsf{tyctors}(D) \in \Sigma_c \quad A = [\cdot, A_1/a_1, \ldots, A_i/a_i]A'}{L : A\, \mathbf{in}\, D(A_1, \ldots, A_i)\, \mathbf{in}\, \Sigma_c} \quad \textsc{Ctor_type}$$

$\boxed{D(a_1, \ldots, a_k) = \{L_1 : \tau_1 | \ldots | L_n : \tau_n\} \in \Sigma_c}$

$$\frac{\mathbf{sort}\, D(a_1, \ldots, a_k) = \{L_1 : \tau_1 | \ldots | L_n : \tau_n\}\, \mathbf{in}\, \Sigma_c}{D(a_1, \ldots, a_k) = \{L_1 : \tau_1 | \ldots | L_n : \tau_n\} \in \Sigma_c} \quad \textsc{DL_sort}$$

$$\frac{\mathbf{type}\, D(a_1, \ldots, a_k) = \{L_1 : A_1 | \ldots | L_n : A_n\}\, \mathbf{in}\, \Sigma_c}{D(a_1, \ldots, a_k) = \{L_1 : A_1 | \ldots | L_n : A_n\} \in \Sigma_c} \quad \textsc{DL_type}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se} \Rightarrow \tau \rightsquigarrow \mathit{ce}}$

$$\frac{x{:}_{\mathit{eff}'}\tau \in \Gamma \quad \mathit{eff}' \leq \mathit{eff}_0}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x \Rightarrow \tau \rightsquigarrow x} \quad \textsc{SurfSyn_var}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{lit} \Rightarrow \mathsf{Int} \rightsquigarrow \mathit{lit}} \quad \textsc{SurfSyn_int}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{blit} \Rightarrow \mathsf{Bool} \rightsquigarrow \mathit{blit}} \quad \textsc{SurfSyn_bool}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se} \Leftarrow \tau \rightsquigarrow \mathit{ce}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se} : \tau \Rightarrow \tau \rightsquigarrow \mathit{ce} : \tau} \quad \textsc{SurfSyn_annot}$$

$$\frac{\begin{array}{c}\mathit{prim}{:}\tau \rightarrow_{\mathit{eff}'} \tau' \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} \mathit{se} \Leftarrow \tau \rightsquigarrow \mathit{ce}\end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{prim}\ \mathit{se} \Rightarrow \tau' \rightsquigarrow \mathit{prim}\ \mathit{ce}} \quad \textsc{SurfSyn_app}$$

$$\frac{\begin{array}{c}f{:}\tau \rightarrow_{\mathit{eff}'} \tau' \in \Sigma_c \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} \mathit{se} \Leftarrow \tau \rightsquigarrow \mathit{ce}\end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} f\ \mathit{se} \Rightarrow \tau' \rightsquigarrow f\ \mathit{ce}} \quad \textsc{SurfSyn_call}$$

$$\frac{\begin{array}{c}A\ \mathsf{eqtype} \\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{se}_1 \Rightarrow A \rightsquigarrow \mathit{ce}_1 \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{se}_2 \Leftarrow A \rightsquigarrow \mathit{ce}_2\end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se}_1 \doteq \mathit{se}_2 \Rightarrow \mathsf{Bool} \rightsquigarrow \mathit{ce}_1 \doteq \mathit{ce}_2} \quad \textsc{SurfSyn_eq}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se} \Leftarrow \tau \rightsquigarrow \mathit{ce}}$

$$\frac{\begin{array}{c}\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{se}_1 \Rightarrow \tau \rightsquigarrow \mathit{ce}_1 \\ \epsilon = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \cdot; \mathit{pat} : \tau \rightarrow [\cdot][[\cdot]]\mathit{se}_2 \Leftarrow_\epsilon \tau' \rightsquigarrow \mathit{ce}_2\end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathsf{let}\, \mathit{pat} = \mathit{se}_1; \mathit{se}_2 \Leftarrow \tau' \rightsquigarrow \mathsf{let}\, y = \mathit{ce}_1; \mathit{ce}_2} \quad \textsc{SurfChk_let}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_1 \Leftarrow \tau_1 \rightsquigarrow ce_1 \quad .. \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_n \Leftarrow \tau_n \rightsquigarrow ce_n}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} (se_1, \,..\,, se_n) \Leftarrow (\tau_1, \,..\,, \tau_n) \rightsquigarrow (ce_1, \,..\,, ce_n)} \quad \textsc{SurfChk_tuple}$$

$$\frac{\begin{array}{l} L : \tau \,\mathbf{in}\, D(\tau_1, \,..\,, \tau_i) \,\mathbf{in}\, \Sigma_c \\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} se \Leftarrow \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} L\, se \Leftarrow D(\tau_1, \,..\,, \tau_i) \rightsquigarrow L\, ce} \quad \textsc{SurfChk_inject}$$

$$\frac{\begin{array}{l} \epsilon = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\epsilon} se \Rightarrow \tau \rightsquigarrow ce' \\ M = \overline{pat{:}\tau, \cdot \to [\cdot][\cdot]\, se} \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \cdot; M \Leftarrow_{\epsilon} \tau' \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{case}\, se\, \mathbf{of}\, \{pat_1 \to se_1 | \,..\, | pat_n \to se_n\} \Leftarrow \tau' \rightsquigarrow \mathsf{let}\, y = ce'; ce} \quad \textsc{SurfChk_case}$$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\mathbf{0}} se_1 \Rightarrow A \rightsquigarrow ce_1 \\ \mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c \\ \Sigma_c; \Gamma \vdash_{\mathbf{1}} z, \cdot; pat : A, \cdot \to [\cdot][[\cdot]] se_2 \Leftarrow_{\mathbf{0}} D(A, B) \rightsquigarrow ce_2 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathbf{1}} \mathbf{iter}\, (pat = se_1)\{se_2\} \Leftarrow B \rightsquigarrow \mathbf{iter}\, (z = ce_1)\{ce_2\}} \quad \textsc{SurfChk_iter}$$

$$\frac{\begin{array}{l} \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} se_1 \Leftarrow \mathsf{Bool} \rightsquigarrow ce_1 \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_2 \Leftarrow \tau \rightsquigarrow ce_2 \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_3 \Leftarrow \tau \rightsquigarrow ce_3 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{if}\, se_1\, \mathbf{then}\, se_2\, \mathbf{else}\, se_3 \Leftarrow \tau \rightsquigarrow \mathbf{if}\, ce_1\, \mathbf{then}\, ce_2\, \mathbf{else}\, ce_3} \quad \textsc{SurfChk_if}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathbf{S}} se \Leftarrow \tau \rightsquigarrow ce}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathbf{return}\, se \Leftarrow \mathsf{Pred}\ \tau \rightsquigarrow \mathbf{return}\, ce} \quad \textsc{SurfChk_return}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathbf{fail} \Leftarrow \mathsf{Pred}\ \tau \rightsquigarrow \mathbf{fail}} \quad \textsc{SurfChk_fail}$$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\mathbf{S}} se_1 \Rightarrow \mathsf{Pred}\ \tau \rightsquigarrow ce_1 \\ \Sigma_c; \Gamma \vdash_{\mathbf{S}} y, \cdot; pat : \tau \to [\cdot][[\cdot]] se_2 \Leftarrow_{\mathbf{S}} \mathsf{Pred}\ \tau' \rightsquigarrow ce_2 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathbf{take}\, pat = se_1; se_2 \Leftarrow \mathsf{Pred}\ \tau' \rightsquigarrow \mathbf{take}\, y = ce_1; ce_2} \quad \textsc{SurfChk_take}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Rightarrow \tau \rightsquigarrow ce}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau \rightsquigarrow ce} \quad \textsc{SurfChk_sub}$$

$$\boxed{\begin{array}{l} \Sigma_s \vdash \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c \vdash \mathit{cprog} \end{array}}$$

$$\frac{\Sigma_c; \cdot \vdash_{\mathit{eff}} se \Leftarrow \tau \rightsquigarrow ce}{\begin{array}{l} \Sigma_s \vdash \mathsf{main} : \tau_{\mathit{eff}} = se \\ \quad \rightsquigarrow \Sigma_c \vdash \mathsf{main} : \tau_{\mathit{eff}} = ce \end{array}} \quad \textsc{Prog_main}$$

$$\frac{\begin{array}{l} M = \overline{pat{:}A, \cdot \to [\cdot][\cdot]\, se} \\ \Sigma_c; \cdot \vdash_{\mathbf{0}} y, \cdot; M \Leftarrow_{\mathbf{0}} B \rightsquigarrow ce \\ \Sigma_s, \mathsf{fun}\ f : A \to_{\mathbf{0}} B = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \vdash \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c, \mathsf{fun}\ f(y : A) \to_{\mathbf{0}} B = ce \vdash \mathit{cprog} \end{array}}{\begin{array}{l} \Sigma_s \vdash \mathsf{fun}\ f : A \to_{\mathbf{0}} B = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \quad \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c \vdash \mathsf{fun}\ f(y : A) \to_{\mathbf{0}} B = ce \quad \mathit{cprog} \end{array}} \quad \textsc{Prog_purefun}$$

$$\frac{\begin{array}{l} M = \overline{pat{:}A, \cdot \to [\cdot][\cdot]\, se} \\ \Sigma_c, f : A \to_{\mathbf{1}} B; \cdot \vdash_{\mathbf{1}} y, \cdot; M \Leftarrow_{\mathbf{0}} B \rightsquigarrow ce \\ \Sigma_s, f : A \to_{\mathbf{1}} B \vdash \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c, f : A \to_{\mathbf{1}} B \vdash \mathit{cprog} \end{array}}{\begin{array}{l} \Sigma_s \vdash \mathsf{fun}\ f : A \to_{\mathbf{1}} B = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \quad \mathit{prog} \\ \quad \rightsquigarrow \Sigma_c \vdash \mathsf{fun}\ f(y : A) \to_{\mathbf{1}} B = ce \quad \mathit{cprog} \end{array}} \quad \textsc{Prog_impurefun}$$

$$\frac{\begin{array}{c} M = \overline{pat{:}\tau, \cdot \to [\cdot][\cdot]\, se} \\ \Sigma_c, f : \tau \to_{\mathbf{S}} \tau'; \cdot \vdash_{\mathbf{S}} y, \cdot; M \Leftarrow_{\mathbf{S}} \tau' \rightsquigarrow ce \\ \Sigma_s, \mathsf{fun}\ f : \tau \to_{\mathbf{S}} \tau' = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \vdash prog \\ \rightsquigarrow \Sigma_c, \mathsf{fun}\ f(y : \tau) \to_{\mathbf{S}} \tau' = ce \vdash cprog \end{array}}{\begin{array}{c} \Sigma_s \vdash \mathsf{fun}\ f : \tau \to_{\mathbf{S}} \tau' = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \quad prog \\ \rightsquigarrow \Sigma_c \vdash \mathsf{fun}\ f(y : \tau) \to_{\mathbf{S}} \tau' = ce \quad cprog \end{array}} \quad \textsc{Prog_specfun}$$

$$\frac{\begin{array}{c} \Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok} \\ \Sigma_s, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} \vdash prog \\ \rightsquigarrow \Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} \vdash cprog \end{array}}{\begin{array}{c} \Sigma_s \vdash \mathbf{sort}\, D(a_1, \ldots, a_i) = \{L_1 : \tau_1 | \ldots | L_n : \tau_n\} prog \\ \rightsquigarrow \Sigma_c \vdash \mathbf{sort}\, D(a_1, \ldots, a_i) = \{L_1 : \tau_1 | \ldots | L_n : \tau_n\} cprog \end{array}} \quad \textsc{Prog_sortdecl}$$

$$\frac{\begin{array}{c} \Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok} \\ \Sigma_s, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} \vdash prog \\ \rightsquigarrow \Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} \vdash cprog \end{array}}{\begin{array}{c} \Sigma_s \vdash \mathbf{type}\, D(a_1, \ldots, a_i) = \{L_1 : A_1 | \ldots | L_n : A_n\} prog \\ \rightsquigarrow \Sigma_c \vdash \mathbf{type}\, D(a_1, \ldots, a_i) = \{L_1 : A_1 | \ldots | L_n : A_n\} cprog \end{array}} \quad \textsc{Prog_typedecl}$$

$\boxed{M\ \mathsf{has_con}}$

$$\frac{}{L\, pat : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_con}} \quad \text{HC_here}$$

$$\frac{M\ \mathsf{has_con}}{\beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_con}} \quad \text{HC_there}$$

$\boxed{M\ \mathsf{has_tup}}$

$$\frac{}{(pat_1, \ldots, pat_n) : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_tup}} \quad \text{HT_here}$$

$$\frac{M\ \mathsf{has_tup}}{\beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_tup}} \quad \text{HT_there}$$

$\boxed{M/var \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}$

$$\frac{}{\cdot/var \overset{y}{\rightsquigarrow}_{\mathit{eff}} \cdot} \quad \text{SV_empty}$$

$$\frac{M/var \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{x : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M/var \overset{y}{\rightsquigarrow}_{\mathit{eff}} \beta \to [\Gamma, x{:}_{\mathit{eff}}\tau][\mathsf{let}\, x = y; \mathcal{E}]se \mid M'} \quad \text{SV_cons}$$

$\boxed{M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}$

$$\frac{}{\cdot/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} \cdot} \quad \text{SC_empty}$$

$$\frac{M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{L\, pat : \tau', \beta \to [\Gamma][\mathcal{E}]se \mid M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} pat : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M'} \quad \text{SC_match}$$

$$\frac{L \neq L' \quad M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{L'\, pat : \tau', \beta \to [\Gamma][\mathcal{E}]se \mid M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'} \quad \text{SC_skip}$$

$$\frac{M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{x : \tau', \beta \to [\Gamma][\mathcal{E}]se \mid M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} z : \tau, \beta \to [\Gamma, x{:}_{\mathit{eff}}\tau'][\mathsf{let}\, x = y; \mathcal{E}]se \mid M'} \quad \text{SC_var}$$

$\boxed{M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}$

$$\frac{}{\cdot/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} \cdot} \quad \text{ET_EMPTY}$$

$$\frac{M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{(pat_1, .., pat_k) : (\tau_1, .., \tau_k), \beta \to [\Gamma][\mathcal{E}]se \mid M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} pat_1 : \tau_1, .., pat_k : \tau_k, \beta \to [\Gamma][\mathcal{E}]se \mid M'} \quad \text{ET_TUPLE}$$

$$\frac{M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{x : (\tau_1, .., \tau_k), \beta \to [\Gamma][\mathcal{E}]se \mid M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} z_1 : \tau_1, .., z_k : \tau_k, \beta \to [\Gamma, x{:}_{\mathit{eff}}(\tau_1, .., \tau_k)][\mathsf{let}\, x = y; \mathcal{E}]se \mid M'} \quad \text{ET_VA}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow ce}$

$$\frac{\Sigma_c; \Gamma, \Gamma' \vdash_{\mathit{eff}_0} se \Leftarrow \tau \rightsquigarrow ce' \quad ce = \mathcal{E}[ce']}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \cdot; \cdot \to [\Gamma'][\mathcal{E}]se \mid M \Leftarrow_\epsilon \tau \rightsquigarrow ce} \quad \text{Cov_DONE}$$

$$\frac{\begin{array}{l} M/var \overset{y}{\rightsquigarrow}_\epsilon M' \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \overline{y}; M' \Leftarrow_\epsilon \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow ce} \quad \text{Cov_VAR}$$

$$\frac{\begin{array}{l} M \ \mathsf{has_con} \\ D(a_1, .., a_k) = \{L_1 : \tau_1' | .. | L_n : \tau_n'\} \in \Sigma_c \\ L_1 : \tau_1'' \,\mathbf{in}\, D(\tau_1, .., \tau_i) \,\mathbf{in}\, \Sigma_c \quad .. \quad L_n : \tau_n'' \,\mathbf{in}\, D(\tau_1, .., \tau_i) \,\mathbf{in}\, \Sigma_c \\ M/L_1{:}\tau_1'' \overset{y}{\rightsquigarrow}_\epsilon M_1 \quad .. \quad M/L_n{:}\tau_n'' \overset{y}{\rightsquigarrow}_\epsilon M_n \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x_1, \overline{y}; M_1 \Leftarrow_\epsilon \tau \rightsquigarrow ce_1 \quad .. \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x_n, \overline{y}; M_n \Leftarrow_\epsilon \tau \rightsquigarrow ce_n \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow \mathbf{case}\, y \,\mathbf{of}\, \{L_1\, x_1 \to ce_1 | .. | L_n\, x_n \to ce_n\}} \quad \text{Cov_CON}$$

$$\frac{\begin{array}{l} M \ \mathsf{has_tup} \\ M/\mathbf{tup} \overset{y}{\rightsquigarrow}_\epsilon M' \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y_1, .., y_n, \overline{y}; M' \Leftarrow_\epsilon \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow \mathsf{let}\,(y_1, .., y_n) = y; ce} \quad \text{Cov_TUP}$$

$\boxed{\mathcal{E}[ce] \rightsquigarrow ce'}$

$$\frac{}{[\cdot][ce] \rightsquigarrow ce} \quad \text{Fill_HOLE}$$

$$\frac{\mathcal{E}[ce] \rightsquigarrow ce'}{\mathsf{let}\, x = ce''; \mathcal{E}[ce] \rightsquigarrow \mathsf{let}\, x = ce''; ce'} \quad \text{Fill_LET}$$

$\boxed{\Sigma_r \vdash \Delta \ \mathsf{wf}}$

$$\frac{}{\Sigma_r \vdash \cdot \ \mathsf{wf}} \quad \text{RWf_EMPTY}$$

$$\frac{\Sigma_r \vdash \Delta \ \mathsf{wf} \quad \mathit{eff} \leq \mathbf{S}}{\Sigma_r \vdash \Delta, x : \tau[\mathit{eff}] \ \mathsf{wf}} \quad \text{RWf_EFF}$$

$$\frac{\Sigma_r \vdash \Delta \ \mathsf{wf} \quad \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{Bool}}{\Sigma_r \vdash \Delta, x : ce[\,\mathsf{log}\,] \ \mathsf{wf}} \quad \text{RWf_LOG}$$

$$\frac{\Sigma_r \vdash \Delta \ \mathsf{wf} \quad \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Rightarrow \mathsf{Pred}\, \tau \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce' \Leftarrow \tau}{\Sigma_r \vdash \Delta, x : ce@ce'[\,\mathsf{res}\,(u)] \ \mathsf{wf}} \quad \text{RWf_RES}$$

$\boxed{\Gamma = |\Delta|}$

$$\frac{}{\cdot = |\cdot|} \quad \textsc{Erase_empty}$$

$$\frac{\Gamma = |\Delta|}{\Gamma, x{:}_{\mathit{eff}}\tau = |\Delta, x : \tau[\mathit{eff}]|} \quad \textsc{Erase_eff}$$

$$\frac{\Gamma = |\Delta|}{\Gamma = |\Delta, x : ce[\,\mathsf{log}\,]|} \quad \textsc{Erase_log}$$

$$\frac{\Gamma = |\Delta|}{\Gamma = |\Delta, x : ce@ce'[\,\mathsf{res}\,(u)]|} \quad \textsc{Erase_res}$$

$\boxed{\Delta' = \mathsf{affinize}(\Delta)}$

$$\frac{}{\cdot = \mathsf{affinize}(\cdot)} \quad \textsc{Aff_empty}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x : \tau[\mathit{eff}] = \mathsf{affinize}(\Delta, x : \tau[\mathit{eff}])} \quad \textsc{Aff_eff}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x : ce[\,\mathsf{log}\,] = \mathsf{affinize}(\Delta, x : ce[\,\mathsf{log}\,])} \quad \textsc{Aff_log}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x : ce@ce'[\,\mathsf{res}\,(0)] = \mathsf{affinize}(\Delta, x : ce@ce'[\,\mathsf{res}\,(0)])} \quad \textsc{Aff_res_used}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x : ce@ce'[\,\mathsf{res}\,(?)] = \mathsf{affinize}(\Delta, x : ce@ce'[\,\mathsf{res}\,(1)])} \quad \textsc{Aff_res_avail}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x : ce@ce'[\,\mathsf{res}\,(?)] = \mathsf{affinize}(\Delta, x : ce@ce'[\,\mathsf{res}\,(?)])} \quad \textsc{Aff_res_opt}$$

$\boxed{\mathsf{zero}(\Delta)}$

$$\frac{}{\mathsf{zero}(\cdot)} \quad \textsc{Zero_empty}$$

$$\frac{\mathsf{zero}(\Delta)}{\mathsf{zero}(\Delta, x : \tau[\mathit{eff}])} \quad \textsc{Zero_eff}$$

$$\frac{\mathsf{zero}(\Delta)}{\mathsf{zero}(\Delta, x : ce[\,\mathsf{log}\,])} \quad \textsc{Zero_log}$$

$$\frac{\mathsf{zero}(\Delta) \quad u \in \{0, ?\}}{\mathsf{zero}(\Delta, x : ce@ce'[\,\mathsf{res}\,(u)])} \quad \textsc{Zero_res}$$

$\boxed{\mathcal{C} = \Delta \Rightarrow \mathcal{C}'}$

$$\frac{}{\mathcal{C} = \cdot \Rightarrow \mathcal{C}} \quad \textsc{Close_empty}$$

$$\frac{\mathcal{C} = \Delta \Rightarrow \forall\, x : \tau.\mathcal{C}'}{\mathcal{C} = \Delta, x : \tau[\mathit{eff}] \Rightarrow \mathcal{C}'} \quad \textsc{Close_eff}$$

$$\frac{\mathcal{C} = \Delta \Rightarrow ce \;\Rightarrow\; \mathcal{C}'}{\mathcal{C} = \Delta, x : ce[\,\mathsf{log}\,] \Rightarrow \mathcal{C}'} \quad \textsc{Close_log}$$

$$\frac{\mathcal{C} = \Delta \Rightarrow \mathcal{C}'}{\mathcal{C} = \Delta, x : ce@ce'[\,\mathsf{res}\,(u)] \Rightarrow \mathcal{C}'} \quad \textsc{Close_res}$$

$\boxed{\Sigma_r; \Gamma \vdash \mathcal{C}\ \mathsf{wf}}$

$$\frac{}{\Sigma_r; \Gamma \vdash \top\ \mathsf{wf}}\ \textsc{CtWf_top}$$

$$\frac{\Sigma_r; \Gamma \vdash \mathcal{C}_1\ \mathsf{wf} \quad \Sigma_r; \Gamma \vdash \mathcal{C}_2\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash \mathcal{C}_1 \wedge \mathcal{C}_2\ \mathsf{wf}}\ \textsc{CtWf_and}$$

$$\frac{\Sigma_r; \Gamma, x{:}_{\mathbf{S}}\tau \vdash \mathcal{C}\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash \forall\, x : \tau.\mathcal{C}\ \mathsf{wf}}\ \textsc{CtWf_forall}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{Bool} \quad \Sigma_r; \Gamma \vdash \mathcal{C}\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash ce \Rightarrow \mathcal{C}\ \mathsf{wf}}\ \textsc{CtWf_impl}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{Bool}}{\Sigma_r; \Gamma \vdash ce\ \mathsf{wf}}\ \textsc{CtWf_atom}$$

$$\frac{}{\Sigma_r; \Gamma \vdash \bot\ \mathsf{wf}}\ \textsc{CtWf_bot}$$

$\boxed{\Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf\ \mathsf{wf}}$

$$\frac{}{\Sigma_r; \Gamma \vdash_{\mathit{eff}} \cdot\ \mathsf{wf}}\ \textsc{PfWf_nil}$$

$$\frac{\begin{array}{l}\mathit{eff}' \in \{\mathbf{S}, \lfloor \mathit{eff} \rfloor\} \\ \Sigma_r; \Gamma, x{:}_{\mathit{eff}'}\tau \vdash_{\mathit{eff}} Pf\ \mathsf{wf}\end{array}}{\Sigma_r; \Gamma \vdash_{\mathit{eff}} x{:}_{\mathit{eff}'}\tau, Pf\ \mathsf{wf}}\ \textsc{PfWf_comp}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{Bool} \quad \Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash_{\mathit{eff}} ce\ [\mathsf{log}], Pf\ \mathsf{wf}}\ \textsc{PfWf_log}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce' \Rightarrow \tau \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{Pred}\ \tau \quad \Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash_{\mathit{eff}} ce@ce'\ [\mathsf{res}], Pf\ \mathsf{wf}}\ \textsc{PfWf_res}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Rightarrow \mathsf{Pred}\ \tau \quad \Sigma_r; \Gamma, y{:}_{\mathbf{S}}\tau \vdash_{\mathit{eff}} Pf\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash_{\mathit{eff}} (y).ce\ [\mathsf{res}], Pf\ \mathsf{wf}}\ \textsc{PfWf_depres}$$

$\boxed{\Sigma_r \vdash RF\ \mathsf{wf}}$

$$\frac{\Sigma_r; \cdot \vdash_{\mathit{eff}} Pf_1\ \mathsf{wf} \quad \Gamma = \mathsf{Bind}(\cdot; Pf_1) \quad \Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf_2\ \mathsf{wf}}{\Sigma_r \vdash Pf_1 \multimap Pf_2\ [\mathit{eff}]\ \mathsf{wf}}\ \textsc{RFWf_rfun}$$

$\boxed{\Sigma_r \vdash f{:}RF}$

$$\frac{f{:}RF \in \Sigma_r}{\Sigma_r \vdash f{:}RF}\ \textsc{RSLk_rfun}$$

$$\frac{f{:}\tau \rightarrow_{\mathit{eff}} \tau' \in \Sigma_r}{\Sigma_r \vdash f{:}x{:}_{\mathit{eff}}\tau, \cdot \multimap y{:}_{\mathit{eff}}\tau', \cdot\ [\mathit{eff}]}\ \textsc{RSLk_sig}$$

$$\frac{f(\mathbf{x} : \tau) \rightarrow_{\mathit{eff}} \tau' = ce \in \Sigma_r}{\Sigma_r \vdash f{:}x{:}_{\mathit{eff}}\tau, \cdot \multimap y{:}_{\mathit{eff}}\tau', \cdot\ [\mathit{eff}]}\ \textsc{RSLk_def}$$

$\boxed{\overline{A} = \{Pf\}}$

$$\frac{}{[\,] = \{\cdot\}}\ \textsc{PfTys_nil}$$

$$\frac{\overline{A} = \{Pf\}}{A{::}\overline{A} = \{x{:}_{\mathbf{0}}A, Pf\}}\ \textsc{PfTys_pure}$$

$$\frac{\overline{A} = \{Pf\}}{\overline{A} = \{x{:}_{\mathbf{S}}\tau, Pf\}} \quad \textsc{PfTys_spec}$$

$$\frac{\overline{A} = \{Pf\}}{\overline{A} = \{ce\ [\mathsf{log}], Pf\}} \quad \textsc{PfTys_log}$$

$$\frac{\overline{A} = \{Pf\}}{\overline{A} = \{ce@ce'\ [\mathsf{res}], Pf\}} \quad \textsc{PfTys_res}$$

$$\frac{\overline{A} = \{Pf\}}{\overline{A} = \{(y).ce\ [\mathsf{res}], Pf\}} \quad \textsc{PfTys_depres}$$

$\boxed{\Sigma_c = \mathsf{Comp}(\Sigma_r)}$

$$\frac{}{\cdot = \mathsf{Comp}(\cdot)} \quad \textsc{RSErase_empty}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r)}{\Sigma_c, f : F = \mathsf{Comp}(\Sigma_r, f : F)} \quad \textsc{RSErase_ext}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r)}{\Sigma_c, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce = \mathsf{Comp}(\Sigma_r, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce)} \quad \textsc{RSErase_fundef}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \overline{A}_1 = \{Pf_1\} \quad A = \mathsf{Prod}(\overline{A}_1) \quad \overline{A}_2 = \{Pf_2\} \quad B = \mathsf{Prod}(\overline{A}_2)}{\Sigma_c, f : A \rightarrow_{\mathit{eff}} B = \mathsf{Comp}(\Sigma_r, f : Pf_1 \multimap Pf_2\ [\mathit{eff}])} \quad \textsc{RSErase_rfun}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r)}{\Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} = \mathsf{Comp}(\Sigma_r, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\})} \quad \textsc{RSErase_sort}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r)}{\Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} = \mathsf{Comp}(\Sigma_r, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\})} \quad \textsc{RSErase_type}$$

$\boxed{\vdash \Sigma_r\ \mathsf{wf}}$

$$\frac{}{\vdash \cdot\ \mathsf{wf}} \quad \textsc{RSWf_empty}$$

$$\frac{\vdash \Sigma_r\ \mathsf{wf}}{\vdash \Sigma_r, f : F\ \mathsf{wf}} \quad \textsc{RSWf_ext}$$

$$\frac{\vdash \Sigma_r\ \mathsf{wf} \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma, x{:}_{\mathit{eff}}\tau \vdash_{\mathit{eff}} ce \Leftarrow \tau'}{\vdash \Sigma_r, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce\ \mathsf{wf}} \quad \textsc{RSWf_fundef}$$

$$\frac{\vdash \Sigma_r\ \mathsf{wf} \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_r \vdash RF\ \mathsf{wf}}{\vdash \Sigma_r, f : RF\ \mathsf{wf}} \quad \textsc{RSWf_rfun}$$

$\boxed{\Sigma_r; \Delta \vdash \mathit{lpf} \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{x{:}ce\ [\mathsf{log}] \in \Delta}{\Sigma_r; \Delta \vdash x \Rightarrow ce \dashv \Delta \rightsquigarrow \top} \quad \textsc{LSyn_var}$$

$$\frac{\begin{array}{l}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad f(\mathbf{x} : \tau) \rightarrow_{\mathit{eff}} \tau' = ce' \in \Sigma_c \quad \mathit{eff} \leq \mathbf{S} \\ \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \tau\end{array}}{\Sigma_r; \Delta \vdash \mathsf{unfold}\ f(ce) \Rightarrow (f\ ce) \doteq [\mathsf{id}(\Gamma), ce/x]ce' \dashv \Delta \rightsquigarrow \top} \quad \textsc{LSyn_unfold}$$

$$\frac{\Sigma_r; \Delta \vdash \mathit{lpf} \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathit{lpf} : ce \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{LSyn_annot}$$

$$\frac{\Sigma_r; \Delta \vdash \mathit{rpf} \Rightarrow \mathbf{return}\ ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathsf{open\text{-}ret}\ \mathit{rpf} \Rightarrow ce \doteq ce' \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{LSyn_openret}$$

$\boxed{\Sigma_r; \Delta \vdash \mathit{lpf} \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{}{\Sigma_r; \Delta \vdash \mathsf{auto} \Leftarrow ce \dashv \Delta \rightsquigarrow ce} \quad \textsc{LChk_auto}$$

$$\frac{\Sigma_r; \Delta \vdash \mathit{lpf} \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathit{lpf} \Leftarrow ce' \dashv \Delta' \rightsquigarrow \mathcal{C} \wedge (ce \Rightarrow (ce'))} \quad \textsc{LChk_sub}$$

$\boxed{\Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}$

$$\frac{x \neq y \quad \Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta, y : \tau[\mathit{eff}] \vdash x : ce@ce' \dashv \Delta', y : \tau[\mathit{eff}]} \quad \textsc{RLk_there_eff}$$

$$\frac{x \neq y \quad \Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta, y : ce''[\mathsf{log}] \vdash x : ce@ce' \dashv \Delta', y : ce''[\mathsf{log}]} \quad \textsc{RLk_there_log}$$

$$\frac{x \neq y \quad \Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta, y : ce''@ce'''[\mathsf{res}\,(u)] \vdash x : ce@ce' \dashv \Delta', y : ce''@ce'''[\mathsf{res}\,(u)]} \quad \textsc{RLk_there_res}$$

$$\frac{u \in \{1, ?\}}{\Sigma_r; \Delta, x : ce@ce'[\mathsf{res}\,(u)] \vdash x : ce@ce' \dashv \Delta, x : ce@ce'[\mathsf{res}\,(0)]} \quad \textsc{RLk_here}$$

$\boxed{\Sigma_r; \Delta \vdash \mathit{rpf} \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta \vdash x \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \top} \quad \textsc{RSyn_var}$$

$$\frac{\Sigma_r; \Delta \vdash \mathit{rpf} \Leftarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathit{rpf} : ce@ce' \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RSyn_annot}$$

$\boxed{\Sigma_r; \Delta \vdash \mathit{rpf} \Leftarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\Sigma_r; \Delta \vdash \mathit{lpf} \Leftarrow ce_1 \doteq ce_2 \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathsf{make\text{-}ret}\ \mathit{lpf} \Leftarrow \mathbf{return}\, ce_1@ce_2 \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RChk_makeret}$$

$$\frac{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow x{:}_{\mathbf{S}}\tau, ce_1@x\ [\mathsf{res}], ce_2@ce_3\ [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathsf{make\text{-}take}\ \mathit{crt} \Leftarrow \mathbf{take}\, x = ce_1; ce_2@ce_3 \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RChk_maketake}$$

$$\frac{\Sigma_r; \Delta \vdash \mathit{rpf} \Rightarrow ce_1@ce_1' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathit{rpf} \Leftarrow ce_2@ce_2' \dashv \Delta' \rightsquigarrow (\mathcal{C} \wedge (ce_1 \doteq ce_2)) \wedge (ce_1' \doteq ce_2')} \quad \textsc{RChk_sub}$$

$\boxed{\Sigma_r; \Delta \vdash \mathit{Pf}_1 = \mathit{Pf}_2 \rightsquigarrow \mathcal{C}}$

$$\frac{}{\Sigma_r; \Delta \vdash \cdot = \cdot \rightsquigarrow \top} \quad \textsc{PfEq_nil}$$

$$\frac{\Sigma_r; \Delta, x : \tau[\mathit{eff}] \vdash \mathit{Pf}_1 = [\mathsf{id}(|\Delta|), x/y]\mathit{Pf}_2 \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash x{:}_{\mathit{eff}}\tau, \mathit{Pf}_1 = y{:}_{\mathit{eff}}\tau, \mathit{Pf}_2 \rightsquigarrow \forall\, x : \tau.\mathcal{C}} \quad \textsc{PfEq_comp}$$

$$\frac{\Sigma_r; \Delta \vdash \mathit{Pf}_1 = \mathit{Pf}_2 \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash ce_1\ [\mathsf{log}], \mathit{Pf}_1 = ce_2\ [\mathsf{log}], \mathit{Pf}_2 \rightsquigarrow ((ce_1 \doteq ce_2)) \wedge \mathcal{C}} \quad \textsc{PfEq_log}$$

$$\frac{\Sigma_r; \Delta \vdash \mathit{Pf}_1 = \mathit{Pf}_2 \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash ce_1@ce_1'\ [\mathsf{res}], \mathit{Pf}_1 = ce_2@ce_2'\ [\mathsf{res}], \mathit{Pf}_2 \rightsquigarrow (((ce_1 \doteq ce_2)) \wedge (ce_1' \doteq ce_2')) \wedge \mathcal{C}} \quad \textsc{PfEq_res}$$

$$\frac{\begin{array}{c}\Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce_1 \Rightarrow \mathsf{Pred}\ \tau \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce_2 \Rightarrow \mathsf{Pred}\ \tau \\ \Sigma_r; \Delta, z : \tau[\mathbf{S}] \vdash [\mathsf{id}(|\Delta|), z/y]\mathit{Pf}_1 = [\mathsf{id}(|\Delta|), z/w]\mathit{Pf}_2 \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash (y).ce_1\ [\mathsf{res}], \mathit{Pf}_1 = (w).ce_2\ [\mathsf{res}], \mathit{Pf}_2 \rightsquigarrow ((ce_1 \doteq ce_2)) \wedge \forall\, z : \tau.\mathcal{C}} \quad \textsc{PfEq_depres}$$

$\boxed{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta}$

$$\frac{}{\Sigma_r \vdash \cdot \sqcap \cdot = \cdot} \quad \textsc{Merge_empty}$$

$$\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta}{\Sigma_r \vdash \Delta_0, x : \tau[\mathit{eff}] \sqcap \Delta_1, x : \tau[\mathit{eff}] = \Delta, x : \tau[\mathit{eff}]} \quad \textsc{Merge_eff}$$

$$\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta}{\Sigma_r \vdash \Delta_0, x : ce[\,\mathsf{log}\,] \sqcap \Delta_1, x : ce[\,\mathsf{log}\,] = \Delta, x : ce[\,\mathsf{log}\,]} \quad \textsc{Merge_log}$$

$$\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta \quad u = u_0 \sqcap u_1}{\Sigma_r \vdash \Delta_0, x : ce@ce'[\,\mathsf{res}\,(u_0)] \sqcap \Delta_1, x : ce@ce'[\,\mathsf{res}\,(u_1)] = \Delta, x : ce@ce'[\,\mathsf{res}\,(u)]} \quad \textsc{Merge_res}$$

$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} q : Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \cdot : \cdot \dashv \Delta \rightsquigarrow \top} \quad \text{RPM_empty}$$

$$\frac{\begin{array}{l}\Gamma = |\Delta| \\ \Sigma_r; \Delta, x : \tau[\mathit{eff}'] \vdash_{\mathit{eff}} q : [\mathsf{id}(\Gamma), x/y]Pf \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} (x, q) : y{:}_{\mathit{eff}'}\tau, Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RPM_comp}$$

$$\frac{\Sigma_r; \Delta, x : ce[\,\mathsf{log}\,] \vdash_{\mathit{eff}} q : Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{log}\ x, q) : ce\ [\mathsf{log}], Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RPM_log}$$

$$\frac{\Sigma_r; \Delta \vdash_{\mathit{eff}} q : Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{log\ auto}, q) : ce\ [\mathsf{log}], Pf \dashv \Delta' \rightsquigarrow\ ce \wedge \mathcal{C}} \quad \text{RPM_auto}$$

$$\frac{\begin{array}{l}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Rightarrow \mathsf{Pred}\ \tau \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} (cpat, (\mathsf{res}\ rpat, q)) : x{:}_{\mathbf{S}}\tau, ce@x\ [\mathsf{res}], Pf \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{do}\ cpat = rpat, q) : (y).ce\ [\mathsf{res}], Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RPM_depres}$$

$$\frac{\Sigma_r; \Delta, x : ce@ce'[\,\mathsf{res}\,(1)] \vdash_{\mathit{eff}} q : Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{res}\ x, q) : ce@ce'\ [\mathsf{res}], Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RPM_res}$$

$$\frac{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{log}\ lpat, q) : ce \doteq ce'\ [\mathsf{log}], Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{res\ return}\ lpat, q) : \mathbf{return}\ ce@ce'\ [\mathsf{res}], Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RPM_return}$$

$$\frac{\begin{array}{l}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce_1 \Rightarrow \mathsf{Pred}\ \tau \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} (cpat, (\mathsf{res}\ rpat_1, (\mathsf{res}\ rpat_2, q))) : x{:}_{\mathbf{S}}\tau, ce_1@x\ [\mathsf{res}], ce_2@ce'\ [\mathsf{res}], Pf \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{res\ take}(cpat, rpat_1); rpat_2, q) : \mathbf{take}\ x = ce_1; ce_2@ce'\ [\mathsf{res}], Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RPM_take}$$

$$\frac{\Sigma_r; \Delta \vdash_{\mathit{eff}} q : Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{res\ fail}, q) : \mathbf{fail}@ce\ [\mathsf{res}], Pf \dashv \Delta' \rightsquigarrow \bot} \quad \text{RPM_fail}$$

$$\frac{\begin{array}{l}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce_1 \Rightarrow \mathsf{Pred}\ \tau \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} (cpat, (\mathsf{log}\ w, (\mathsf{res}\ rpat, q))) : x{:}_{\mathbf{S}}\tau, x \doteq ce_1\ [\mathsf{log}], ce_2@ce'\ [\mathsf{res}], Pf \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{res\ letw}]\ cpat; rpat, q) : \mathsf{let}\ x = ce_1; ce_2@ce'\ [\mathsf{res}], Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RPM_rlet}$$

$$\frac{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{res}\ rpat, q) : ce_1@ce'\ [\mathsf{res}], Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{res\ iftrue}; rpat, q) : \mathbf{if}\ ce\ \mathbf{then}\ ce_1\ \mathbf{else}\ ce_2@ce'\ [\mathsf{res}], Pf \dashv \Delta' \rightsquigarrow\ ce \wedge \mathcal{C}} \quad \text{RPM_iftrue}$$

$$\frac{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{res}\ rpat, q) : ce_2@ce'\ [\mathsf{res}], Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{res\ iffalse}; rpat, q) : \mathbf{if}\ ce\ \mathbf{then}\ ce_1\ \mathbf{else}\ ce_2@ce'\ [\mathsf{res}], Pf \dashv \Delta' \rightsquigarrow\ (\mathbf{Not}\ ce) \wedge \mathcal{C}} \quad \text{RPM_iffalse}$$

$$\frac{\begin{array}{l}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Gamma = |\Delta| \quad \mathit{eff}' = \lfloor \mathit{eff} \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{ce} \Rightarrow D(\tau_1, .., \tau_i) \\ L{:}\tau \in \mathrm{ctors}(D) \in \Sigma_c \\ \mathit{ce}'' = \mathsf{branch}(L, x, \mathit{ce}''') \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathit{cpat}, (\mathsf{log}\ \mathit{lpat}, (\mathsf{res}\ \mathit{rpat}, q))) : x{:}_{\mathbf{S}}\tau, (L\,x) \doteq \mathit{ce}\ [\mathsf{log}], \mathit{ce}''@\mathit{ce}'\ [\mathsf{res}], \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{res\ case}\ L(\mathit{cpat}); \mathit{lpat}; \mathit{rpat}, q) : \mathit{ce}'''@\mathit{ce}'\ [\mathsf{res}], \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathsf{is}\ \mathit{ce}\ L \wedge \mathcal{C}} \quad \text{RPM_CASE}$$

$$\frac{\begin{array}{l}f(\mathbf{x} : \tau) \rightarrow_{\mathit{eff}'} \tau' = \mathit{ce}_0 \in \Sigma_r \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{res}\ \mathit{rpat}, q) : ([\cdot, \mathit{ce}/x]\mathit{ce}_0)@\mathit{ce}'\ [\mathsf{res}], \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{res\ unfold}; \mathit{rpat}, q) : f\ \mathit{ce}@\mathit{ce}'\ [\mathsf{res}], \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RPM_UNFOLD}$$

$$\frac{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{res}\ \mathit{rpat}, q) : \mathit{ce}@\mathit{ce}'\ [\mathsf{res}], \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} (\mathsf{res\ annot}; \mathit{rpat}, q) : (\mathit{ce} : \tau)@\mathit{ce}'\ [\mathsf{res}], \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RPM_ANNOT}$$

$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Rightarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\begin{array}{l}\Gamma = |\Delta| \quad \Sigma_r; \Gamma \vdash_{\mathit{eff}} \mathit{Pf}\ \mathsf{wf} \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{crt} : \mathit{Pf} \Rightarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RCSyn_annot}$$

$$\frac{\begin{array}{l}\Sigma_r \vdash f{:}\mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathit{eff}'] \quad \mathit{eff}' \leq \mathit{eff} \quad \mathit{eff}'' = \lfloor \mathit{eff} \rfloor \\ \Sigma_r; \Delta \vdash_{\mathit{eff}''} \mathit{rsp} : \mathit{Pf}_1 \multimap \mathit{Pf}_2 \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} f\ \mathit{rsp} \Rightarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RCSyn_call}$$

$$\frac{\begin{array}{l}\mathit{prim}{:}\mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathit{eff}'] \quad \mathit{eff}' \leq \mathit{eff} \quad \mathit{eff}'' = \lfloor \mathit{eff} \rfloor \\ \Sigma_r; \Delta \vdash_{\mathit{eff}''} \mathit{rsp} : \mathit{Pf}_1 \multimap \mathit{Pf}_2 \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{prim}\ \mathit{rsp} \Rightarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RCSyn_primapp}$$

$$\frac{\Sigma_r; \Delta \vdash \mathit{rpf} \Rightarrow \mathbf{take}\, x = \mathit{ce}_1; \mathit{ce}_2@\mathit{ce}_3 \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathsf{open\text{-}take}\ \mathit{rpf} \Rightarrow x{:}_{\mathbf{S}}\tau, \mathit{ce}_1@x\ [\mathsf{res}], \mathit{ce}_2@\mathit{ce}_3\ [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RCSyn_opentake}$$

$$\frac{\begin{array}{l}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c \\ \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathit{ce} \Rightarrow \mathsf{Pred}\ D(A, B) \\ \Sigma_r; \Delta \vdash_{\mathbf{0}} \mathit{crt}_1 \Leftarrow x{:}_{\mathbf{0}}A, \mathit{ce}@\mathit{Next}\ x\ [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta', x : A[\mathbf{0}], y : \mathit{ce}@\mathit{Next}\ x[\mathsf{res}\,(1)] \vdash_{\mathbf{1}} \mathit{crt}_2 \Leftarrow z{:}_{\mathbf{0}}D(A, B), \mathit{ce}@z\ [\mathsf{res}], \cdot \dashv \Delta', x : A[\mathbf{0}], y : \mathit{ce}@\mathit{Next}\ x[\mathsf{res}\,(0)] \sim\end{array}}{\Sigma_r; \Delta \vdash_{\mathbf{1}} \mathbf{iter}\,[\mathit{ce}](q = \mathit{crt}_1)\{\mathit{crt}_2\} \Rightarrow z{:}_{\mathbf{0}}B, \mathit{ce}@\mathit{Done}\ z\ [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C} \wedge \forall x : A.\mathcal{C}'}$$

$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} y{:}\mathit{ce}\ \mathsf{is}\ L\,x \rightarrow \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\begin{array}{l}\Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} \mathit{ce} \Rightarrow D(\tau_1, .., \tau_i) \\ L : \tau\ \mathbf{in}\ D(\tau_1, .., \tau_i)\ \mathbf{in}\ \Sigma_c \\ \Sigma_r; \Delta, x : \tau[\mathit{eff}], y : \mathit{ce} \doteq L\,x[\mathsf{log}] \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta', x : \tau[\mathit{eff}], y : \mathit{ce} \doteq L\,x[\mathsf{log}] \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} y{:}\mathit{ce}\ \mathsf{is}\ L\,x \rightarrow \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \forall x : \tau.\mathit{ce} \doteq L\,x \Rightarrow \mathcal{C}} \quad \text{CBr_branch}$$

$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\begin{array}{l}\Sigma_r; \Delta_0 \vdash_{\mathit{eff}} \mathit{crt}_1 \Rightarrow \mathit{Pf}' \dashv \Delta_1 \rightsquigarrow \mathcal{C} \\ \mathit{eff}' = \lfloor \mathit{eff} \rfloor \\ \Sigma_r; \Delta_1 \vdash_{\mathit{eff}'} q : \mathit{Pf}' \dashv \Delta_2 \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta_2 \vdash_{\mathit{eff}} \mathit{crt}_2 \Leftarrow \mathit{Pf} \dashv \Delta_3, \Delta_4 \rightsquigarrow \mathcal{C}_2 \\ \mathsf{len}(\Delta_1) = \mathsf{len}(\Delta_3) \\ \mathsf{zero}(\Delta_4) \\ \mathcal{C}' = \Delta_4 \Rightarrow (\mathcal{C}_1 \wedge \mathcal{C}_2)\end{array}}{\Sigma_r; \Delta_0 \vdash_{\mathit{eff}} \mathsf{let}\ q = \mathit{crt}_1; \mathit{crt}_2 \Leftarrow \mathit{Pf} \dashv \Delta_3 \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RCChk_let}$$

$$\frac{\begin{array}{l}\Sigma_r; \Delta \vdash \mathit{lpf} \Rightarrow \mathit{ce} \dashv \Delta_1 \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta_1, x : \mathit{ce}[\,\mathsf{log}\,] \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta_2, x : \mathit{ce}[\,\mathsf{log}\,] \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathsf{let\,log}\, x = \mathit{lpf}; \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta_2 \rightsquigarrow \mathcal{C} \wedge (\mathit{ce} \Rightarrow \mathcal{C}')} \quad \text{RCChk_letlog}$$

$$\frac{\begin{array}{l}\Sigma_r; \Delta \vdash \mathit{rpf} \Rightarrow \mathit{ce}@\mathit{ce}' \dashv \Delta_1 \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta_1, x : \mathit{ce}@\mathit{ce}'[\,\mathsf{res}\,(1)] \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta_2, x : \mathit{ce}@\mathit{ce}'[\,\mathsf{res}\,(u)] \rightsquigarrow \mathcal{C}' \\ u \in \{0, ?\}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathsf{let\,res}\, x = \mathit{rpf}; \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta_2 \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RCChk_letres}$$

$$\frac{\begin{array}{l}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Gamma = |\Delta| \quad \mathit{eff}' = \lfloor \mathit{eff} \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{ce} \Rightarrow \tau \\ \Sigma_r; \Delta, x : \tau[\mathit{eff}'], w : x \doteq \mathit{ce}[\,\mathsf{log}\,] \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta_1, x : \tau[\mathit{eff}'], w : x \doteq \mathit{ce}[\,\mathsf{log}\,] \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathsf{let\,core}\,[w]x = \mathit{ce}; \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta_1 \rightsquigarrow \forall\, x : \tau.(x \doteq \mathit{ce}) \Rightarrow \mathcal{C}} \quad \text{RCChk_letcore}$$

$$\frac{\begin{array}{l}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Gamma = |\Delta| \quad \mathit{eff}' = \lfloor \mathit{eff} \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{ce} \Rightarrow (\tau_1, \ldots, \tau_n) \\ \Delta' = x_1 : \tau_1[\mathit{eff}'], \ldots, x_n : \tau_n[\mathit{eff}'], w : (x_1, \ldots, x_n) \doteq \mathit{ce}[\,\mathsf{log}\,] \\ \Sigma_r; \Delta, \Delta' \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta_1, \Delta' \rightsquigarrow \mathcal{C} \\ \mathcal{C}' = \Delta' \Rightarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathsf{let\,core}\,[w](x_1, \ldots, x_n) = \mathit{ce}; \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta_1 \rightsquigarrow \mathcal{C}'} \quad \text{RCChk_letcoretup}$$

$$\frac{\begin{array}{l}\mathit{eff}' = \lfloor \mathit{eff} \rfloor \\ \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{ce} \Leftarrow \mathsf{Bool} \\ \Sigma_r; \Delta, x : \mathit{ce} \doteq \mathsf{true}[\,\mathsf{log}\,] \vdash_{\mathit{eff}'} \mathit{crt}_1 \Leftarrow \mathit{Pf} \dashv \Delta_1, x : \mathit{ce} \doteq \mathsf{true}[\,\mathsf{log}\,] \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta, x : \mathit{ce} \doteq \mathsf{false}[\,\mathsf{log}\,] \vdash_{\mathit{eff}'} \mathit{crt}_2 \Leftarrow \mathit{Pf} \dashv \Delta_2, x : \mathit{ce} \doteq \mathsf{false}[\,\mathsf{log}\,] \rightsquigarrow \mathcal{C}_2 \\ \Sigma_r \vdash \Delta_1 \sqcap \Delta_2 = \Delta'\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathbf{if}\,[x]\mathit{ce}\,\mathbf{then}\,\mathit{crt}_1\,\mathbf{else}\,\mathit{crt}_2 \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow (\mathit{ce} \doteq \mathsf{true} \Rightarrow \mathcal{C}_1) \wedge (\mathit{ce} \doteq \mathsf{false} \Rightarrow \mathcal{C}_2)} \quad \text{RCChk_if}$$

$$\frac{\begin{array}{l}\mathit{eff}' = \lfloor \mathit{eff} \rfloor \\ \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{ce} \Rightarrow D(\tau_1, \ldots, \tau_i) \\ D(a_1, \ldots, a_k) = \{L_1 : \tau_1' | \ldots | L_n : \tau_n'\} \in \Sigma_c \\ \Sigma_r; \Delta \vdash_{\mathit{eff}'} y{:}\mathit{ce}\ \mathsf{is}\ L_1\, x_1 \rightarrow \mathit{crt}_1 \Leftarrow \mathit{Pf} \dashv \Delta_1 \rightsquigarrow \mathcal{C}_1 \quad \ldots \quad \Sigma_r; \Delta \vdash_{\mathit{eff}'} y{:}\mathit{ce}\ \mathsf{is}\ L_n\, x_n \rightarrow \mathit{crt}_n \Leftarrow \mathit{Pf} \dashv \Delta_n \rightsquigarrow \mathcal{C}_n \\ \Sigma_r \vdash \sqcap(\Delta_1; \ldots; \Delta_n) = \Delta'\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathbf{case}\,[y]\mathit{ce}\,\mathbf{of}\,\{L_1\, x_1 \rightarrow \mathit{crt}_1 | \ldots | L_n\, x_n \rightarrow \mathit{crt}_n\} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}_1 \wedge \mathcal{C}_n} \quad \text{RC}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathsf{exfalso} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \bot} \quad \text{RCChk_exfalso}$$

$$\frac{\begin{array}{l}\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Rightarrow \mathit{Pf}' \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash \mathit{Pf}' = \mathit{Pf} \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RCChk_sub}$$

$$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : \mathit{Pf}_1 \multimap \mathit{Pf}_2 \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}}$$

$$\frac{}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \cdot : \cdot \multimap \mathit{Pf} \gg \mathit{Pf} \dashv \Delta \rightsquigarrow \top} \quad \text{Spine_nil}$$

$$\frac{\begin{array}{l}\Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} \mathit{ce} \Leftarrow \tau \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : [\mathsf{id}(|\Delta|), \mathit{ce}/x]\mathit{Pf}_1 \multimap [\mathsf{id}(|\Delta|), \mathit{ce}/x]\mathit{Pf}_2 \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{ce}, \mathit{rsp} : x{:}_{\mathbf{0}}\tau, \mathit{Pf}_1 \multimap \mathit{Pf}_2 \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{Spine_pure}$$

$$\frac{\begin{array}{l}\Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathit{ce} \Leftarrow \tau \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : [\mathsf{id}(|\Delta|), \mathit{ce}/x]\mathit{Pf}_1 \multimap [\mathsf{id}(|\Delta|), \mathit{ce}/x]\mathit{Pf}_2 \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{ce}, \mathit{rsp} : x{:}_{\mathbf{S}}\tau, \mathit{Pf}_1 \multimap \mathit{Pf}_2 \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{Spine_spec}$$

$$\frac{\begin{array}{c}\Sigma_r; \Delta \vdash lpf \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} rsp : Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} lpf, rsp : ce\ [\mathsf{log}], Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Spine_log}$$

$$\frac{\begin{array}{c}\Sigma_r; \Delta \vdash rpf \Leftarrow ce_1 @ ce_2 \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} rsp : Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} rpf, rsp : ce_1 @ ce_2\ [\mathsf{res}], Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Spine_res}$$

$$\frac{\begin{array}{c}\Sigma_r; \Delta \vdash rpf \Rightarrow ce' @ ce'' \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} rsp : [\mathsf{id}(|\Delta'|), ce''/y]Pf_1 \multimap [\mathsf{id}(|\Delta'|), ce''/y]Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} rpf, rsp : (y).ce\ [\mathsf{res}], Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow (\mathcal{C} \wedge (ce \doteq ce')) \wedge \mathcal{C}'} \quad \textsc{Spine_depres}$$

$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} rsp : Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \cdot : \cdot \dashv \Delta \rightsquigarrow \top} \quad \textsc{Tup_nil}$$

$$\frac{\begin{array}{c}\Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} ce \Leftarrow \tau \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} rsp : [\mathsf{id}(|\Delta|), ce/x]Pf \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} ce, rsp : x{:}_{\mathit{eff}}\tau, Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{Tup_core}$$

$$\frac{\begin{array}{c}\Sigma_r; \Delta \vdash lpf \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} rsp : Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} lpf, rsp : ce\ [\mathsf{log}], Pf \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Tup_log}$$

$$\frac{\begin{array}{c}\Sigma_r; \Delta \vdash rpf \Leftarrow ce @ ce' \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} rsp : Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} rpf, rsp : ce @ ce'\ [\mathsf{res}], Pf \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Tup_res}$$

$$\frac{\begin{array}{c}\Sigma_r; \Delta \vdash rpf \Rightarrow ce' @ ce'' \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} rsp : [\mathsf{id}(|\Delta'|), ce''/y]Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} rpf, rsp : (y).ce\ [\mathsf{res}], Pf \dashv \Delta'' \rightsquigarrow (\mathcal{C} \wedge (ce \doteq ce')) \wedge \mathcal{C}'} \quad \textsc{Tup_depres}$$

$\boxed{\Sigma_r \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}}$

$$\frac{\Sigma_r; \cdot \vdash_{\mathit{eff}} crt \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r \vdash \mathsf{main} : Pf_{\mathit{eff}} = crt \dashv \Sigma_r \rightsquigarrow \mathcal{C}} \quad \textsc{RProg_main}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok} \\ \Sigma_r, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r \vdash \mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}} \quad \textsc{RProg_sortdecl}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok} \\ \Sigma_r, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r \vdash \mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 | .. | L_n : A_n\} rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}} \quad \textsc{RProg_typedecl}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \cdot, x{:}_{\mathbf{0}} A \vdash_{\mathbf{0}} ce \Leftarrow B \\ \Sigma_r, \mathsf{fun}\ f(x : A) \rightarrow_{\mathbf{0}} B = ce \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(x : A) \rightarrow_{\mathbf{0}} B = ce \quad rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}} \quad \textsc{RProg_purefun}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c, f : A \rightarrow_{\mathbf{1}} B; \cdot, x{:}_{\mathbf{0}} A \vdash_{\mathbf{1}} ce \Leftarrow B \\ \Sigma_r, f : A \rightarrow_{\mathbf{1}} B \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(x : A) \rightarrow_{\mathbf{1}} B = ce \quad rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}} \quad \textsc{RProg_impurefun}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c, f : \tau \rightarrow_{\mathbf{S}} \tau'; \cdot, x{:}_{\mathbf{S}} \tau \vdash_{\mathbf{S}} ce \Leftarrow \tau' \\ \Sigma_r, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathbf{S}} \tau' = ce \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathbf{S}} \tau' = ce \quad rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}} \quad \textsc{RProg_specfun}$$

$$\frac{\begin{array}{l}\Sigma_r \vdash Pf_1 \multimap Pf_2\ [\mathbf{0}]\ \mathsf{wf} \\ \Sigma_r; \cdot \vdash_{\mathbf{0}} q : Pf_1 \dashv \Delta \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta \vdash_{\mathbf{0}} crt \Leftarrow Pf_2 \dashv \Delta' \rightsquigarrow \mathcal{C}_2 \\ \mathsf{zero}(\Delta') \\ \mathcal{C} = \Delta' \Rightarrow (\mathcal{C}_1 \wedge \mathcal{C}_2) \\ \Sigma_r, f : Pf_1 \multimap Pf_2\ [\mathbf{0}] \vdash rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(q) : Pf_1 \multimap Pf_2\ [\mathbf{0}] = crt \quad rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RProg_purerefined}$$

$$\frac{\begin{array}{l}\Sigma_r \vdash Pf_1 \multimap Pf_2\ [\mathbf{1}]\ \mathsf{wf} \\ \Sigma_r; \cdot \vdash_{\mathbf{0}} q : Pf_1 \dashv \Delta \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r, f : Pf_1 \multimap Pf_2\ [\mathbf{1}]; \Delta \vdash_{\mathbf{1}} crt \Leftarrow Pf_2 \dashv \Delta' \rightsquigarrow \mathcal{C}_2 \\ \mathsf{zero}(\Delta') \\ \mathcal{C} = \Delta' \Rightarrow (\mathcal{C}_1 \wedge \mathcal{C}_2) \\ \Sigma_r, f : Pf_1 \multimap Pf_2\ [\mathbf{1}] \vdash rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(q) : Pf_1 \multimap Pf_2\ [\mathbf{1}] = crt \quad rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RProg_impurerefined}$$

$$\frac{\begin{array}{l}\Sigma_r \vdash Pf_1 \multimap Pf_2\ [\mathbf{S}]\ \mathsf{wf} \\ \Sigma_r; \cdot \vdash_{\mathbf{S}} q : Pf_1 \dashv \Delta \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r, f : Pf_1 \multimap Pf_2\ [\mathbf{S}]; \Delta \vdash_{\mathbf{S}} crt \Leftarrow Pf_2 \dashv \Delta' \rightsquigarrow \mathcal{C}_2 \\ \mathsf{zero}(\Delta') \\ \mathcal{C} = \Delta' \Rightarrow (\mathcal{C}_1 \wedge \mathcal{C}_2) \\ \Sigma_r, f : Pf_1 \multimap Pf_2\ [\mathbf{S}] \vdash rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(q) : Pf_1 \multimap Pf_2\ [\mathbf{S}] = crt \quad rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RProg_specrefined}$$

$\boxed{\Sigma_r \vdash \sqcap\overline{\Delta} = \Delta}$

$$\frac{}{\Sigma_r \vdash \sqcap\Delta_0; \cdot = \Delta_0} \quad \text{NMerge_base}$$

$$\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta' \quad \Sigma_r \vdash \sqcap(\Delta'; \overline{\Delta}) = \Delta}{\Sigma_r \vdash \sqcap(\Delta_0; \Delta_1; \overline{\Delta}) = \Delta} \quad \text{NMerge_step}$$

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Definition rules:        292 good    0 bad
Definition rule clauses: 651 good    0 bad
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